An empirical method for studying AI-generated mathematics

Safal R. Aryal

October 18, 2026

Abstract

I present a method for analyzing large collections of formally verified mathematical proofs (e.g. OpenAI's) grounded in some historical and philosophical musings on mathematics in the age of AI. Many of the quantities involved, such as the size of a proof term or the number of steps that do no work, are already computed by work on improving automated provers, where they serve as targets. My aim is to turn them into measurements of proofs themselves: compared against a reference library and against each proof's own cut-free form, checked against theorems they must obey, and controlled for how the proofs were compiled. With these instruments, I measure how (much) machine-generated mathematics diverges from human mathematics, and find that AI-written formal proofs realize far less of the logical compression available to them than human mathematicians do (31% against 73% in mathlib, and 55% against 86% on proofs that call no automation). These figures are broadly in line with reports from those who build provers and proof-improvement tools that AI-written proofs are often longer than they need to be.

1 Introduction

Rather, as we do not ask what hope of gain makes a little bird warble, since we know that it takes delight in singing because it is for that very singing that the bird was made, so there is no need to ask why the human mind undertakes such toil in seeking out these secrets of the heavens. For the reason why the mind was joined to the senses by our Maker is not only so that Man should maintain himself, which many species of living things can do far more cleverly with the aid of even an irrational mind, but also so that from those things which we perceive with our eyes to exist we should strive towards the causes of their being and becoming, although we should get nothing else useful from them. And just as other animals, and the human body, are sustained by food and drink, so the very spirit of Man, which is something distinct from Man, is nourished,

is increased, and in a sense grows up on this diet of knowledge, and is more like the dead than the living if it is touched by no desire for these things. Therefore as by the providence of Nature nourishment is never lacking for living things, so we can say with justice that the reason why there is such great variety in things, and treasures so well concealed in the fabric of the heavens, is so that fresh nourishment should never be lacking for the human mind, and it should never disdain it as stale, nor be inactive, but should have in this universe an inexhaustible workshop in which to busy itself.

— Johannes Kepler

1.1 Up to the stars

Contemporary AI's seeming resolution of the Navier–Stokes problem *(OpenAI 2026a)* confirms that machine learning can now generate complex mathematical proofs at an unprecedented scale, forcing a critical reflection on the nature of proof itself. Such reflections (especially those produced in the academy) customarily began with discussions of the foundational crisis that mathematics underwent in the late 19th and early 20th centuries. But while this was a pivotal chapter in the history of mathematics (and arguably the most major moment for proof since Euclid compiled the *Elements (Euclid 1908)*), I do not believe that this crisis accurately captures the epistemological and methodological challenges presented by AI-generated proofs. To fully understand this moment in history, I believe that one has to look back further: to Tycho Brahe and Johannes Kepler's collaborative and sequential efforts to collect astronomical data during the Scientific Revolution.

For years, Brahe labored at his observatories on the island of Hven to meticulously record decades of naked-eye planetary and stellar positions using large, custom-built pre-telescopic instruments (e.g. he used, among many other instruments, a massive brass quarter-circle arc mounted to a solid north-south wall with a 2-meter radius, used to measure stellar altitudes as stars crossed the meridian) *(Brahe 1598; Thoren 1990)*. When Kepler was appointed as Brahe's successor as the Imperial Mathematician to Holy Roman Emperor Rudolf II, he inherited this vast and unprecedentedly accurate observational archive, using the empirical measurements to mathematically test and formulate his laws of planetary motion *(Kepler 1609; Kepler 1992)*.

Prior to his death, Tycho Brahe had been puzzled by the motion of Mars: he was not able to find a mathematically parsimonious description of how the planet moved that also minimized measurement error. To solve Brahe's puzzle, Kepler had to make five key evidentiary innovations. First, he was able to formulate a mathematical method that made calculations possible that used the position of the true sun (the actual sun you see moving across the sky), rather than the mean sun (an imaginary, average sun that moves at a

completely steady speed all year–a favorite abstraction of the astronomers of antiquity). This allowed for a more accurate view of the relative motions of the Sun, Earth, and Mars, eliminating many longitudinal errors that existed in Brahe's interpretations of the data.

Kepler's second innovation concerned the bisection of eccentricity for the Earth–Sun orbit, which helped reduce the discrepancies in Mars' predicted position from over 4 degrees to less than an arcminute. This refinement was made possible in part by his third innovation, the area rule: a line joining the Sun and a planet sweeps out equal areas in equal periods of time. Together with the use of the true sun, the area rule allowed Kepler to calculate planetary positions with remarkable accuracy. Moreover, it constrained the possible orbital geometries: given the temporal and positional constraints imposed by Brahe's observations, Kepler found that the ellipse provided the most accurate mathematical description of Mars' motion *(Stephenson 1987)*.

None of this, of course, would have been possible without Brahe's dataset. Brahe's data made persistent discrepancies available for mathematical investigation at an unprecedented level of precision. Kepler was able to use these discrepancies to revise established assumptions about planetary motion, developing a system in which geometrical constructions were increasingly informed by physical considerations. This represented a fundamental change in the conception of astronomy, from the mathematical description of celestial positions toward the study of physical bodies and their motions. Better known than Brahe or Kepler, Galileo's telescopic observations further advanced this transition: lunar mountains, sunspots, the phases of Venus, and Jupiter's satellites challenged important components of Scholastic cosmology and expanded the empirical scope of astronomical inquiry *(Galilei 1989; Galilei and Scheiner 2010)*. However, while Kepler had developed remarkably accurate geometrical models and Galileo had established new observational facts about the heavens, a unified physical theory of terrestrial and celestial motion remained an open problem.

The resolution of this problem, of course, came from Isaac Newton (beginning in *De motu corporum in gyrum (Whiteside 1974)* and culminating in the famous *Philosophiæ Naturalis Principia Mathematica (Newton 1687; Newton 1999)*). *De motu* (which our discussion here will focus on) demonstrated how mathematical theory could strengthen the explanatory standing of an astronomical system even when competing models accounted for the available observations with comparable accuracy. By this point, the quality and resolution of observational evidence were already considerable, largely due to the work of Brahe and his successors. The remaining problem was to establish a more compelling relationship between the observed motions of celestial bodies and the principles of earthly mechanics. Newton's innovation in *De motu* was to show that the mathematical structures underlying these two domains could be systematically related, thereby providing a stronger basis for preferring the Keplerian system over its competitors.

The significance of Newton's work was that it established precise mathematical relationships between theories that had previously developed somewhat independently. By demonstrating how centripetal forces could account for Kepler's laws, and deriving further consequences for planetary motion, Newton substantially increased the explanatory power of the laws of Kepler's system. These deductions also provided a form of evidential support: a mathematical relationship between independently motivated theories could increase the credibility of both, while opening up avenues for further investigation. The resulting framework was therefore more robust than its predecessors, insofar as it could account for a wider range of phenomena through a common set of principles; a process which culminated in the development of the calculus and Newton's laws of motion.

Could the laws of motion have been discovered without these innovations in turning observations into data, and data into evidence? Perhaps in some possible world they could have been, but in the world we inhabit, I would wager they could not. Astronomy had long been treated primarily as a geometrical discipline, in which the principal task was to formulate mathematically coherent descriptions of celestial motion. Brahe's systematic and unusually precise observations made it possible to identify discrepancies at a resolution that earlier astronomical models could not adequately accommodate, while Kepler's willingness to revise the assumptions through which these observations were interpreted led to a substantially more accurate mathematical account of planetary motion. Newton's achievement depended on this prior work, since the integration of celestial and terrestrial mechanics required a sufficiently accurate mathematical description of orbital motion to establish a meaningful relationship between the two. The transition to Newtonian mechanics thus depended on a methodological transformation in which observations were systematically collected, discrepancies were treated as grounds for revising existing models, and mathematical relationships were used to establish connections that the observations alone could not have revealed. Without these developments, the data necessary for Newton's synthesis might have existed in some form, but the evidentiary and mathematical structure through which they became scientifically consequential would have been considerably more difficult to establish.

I now turn to argue that mathematics – the purest, most pristine, deductive science – is undergoing a similar transformation. Notwithstanding significant social and economic challenges, we have massive amounts of mathematical data, which we may turn into mathematical evidence to test mathematical claims.

This paper is a small attempt at doing so. I compiled 96 projects from OpenAI's library of AI-written mathematics, three smaller AI-written corpora, and two human research projects against the same version of mathlib, Lean's main library, and measured the proof terms that Lean checks. The structure of proofs has rarely been measured empirically, let alone at this scale, so part of the exercise is constructing a methodology

strong enough to put numbers to it. Many of the numbers themselves are not new: the people who build and improve automated provers already compute them, as targets to optimize (see "What is already measured"). What is new here is the attempt to use them as measurements of proofs. The measures are checked against theorems they must obey (where such theorems exist), against how much mathlib differs from itself, and against controls for how the corpora were compiled, and I report where results moved along the way. Three results hold up under these controls. First, the mix of steps in OpenAI's proofs is 5.8 times as far from mathlib's as random samples of mathlib are from the rest of mathlib, and still 3.4 to 3.9 times as far once the output of automated tactics is collapsed (though this comparison does not control for subject matter). Second, OpenAI's proofs state many intermediate claims that do no work: 40% of them are larger with their intermediate claims than without them, against 13% of mathlib's. Third, they realize less of the compression available to them (the amount by which a proof could be shortened by proving each repeated piece once and reusing it): 31% against 73% in mathlib, and 55% against 86% on proofs that call no automation. Other first impressions did not hold up. OpenAI's proofs are longer mostly because they call more automation, and once automation is set aside they are no more compressible than mathlib's.

1.2 The "foundational crisis" of the 19th and 20th centuries

> *Already at this time I should like to assert what the final outcome will be: mathematics is a presuppositionless science. To found it I do not need God, as does Kronecker, or the assumption of a special faculty of our understanding attuned to the principle of mathematical induction, as does Poincaré, or the primal intuition of Brouwer, or, finally, as do Russell and Whitehead, axioms of infinity, reducibility, or completeness, which in fact are actual, contentual assumptions that cannot be compensated for by consistency proofs.*
>
> — David Hilbert, *Die Grundlagen der Mathematik (Hilbert 1928; Hilbert 1967)*

What is mathematics? We can separate this query into two parts: one *metaphysical* and one *epistemic*. The former asks what mathematical objects (and mathematics itself) are, while the latter concerns our mathematical *knowledge*. While we now conceptualize these as two separate questions, this has not always been the case. The contemplation of such questions reached a new height in the so-called "foundational crisis" mathematics faced in the late 19th and early 20th centuries. We will begin our review of this crisis by comparing views held on this question by two bitter rivals: the proto-constructivist Leopold Kronecker, and (putative) father of set theory, Georg Cantor. Consider the famous saying commonly attributed to Kronecker:

God made the integers, all else is the work of man. (Weber 1893, p. 19)

Kronecker's mathematical work reflected this position: in number theory and algebra, he sought to ground mathematical constructions in the integers and finite arithmetic operations, resisting mathematical objects whose existence depended on completed infinite processes. Cantor pursued a sharply different program, developing set theory and the transfinite numbers, which treated infinite collections as legitimate mathematical objects in their own right, pursuant to the following maxim:

> *Mathematics is completely free in its development and is only bound to the self-evident consideration that its concepts are both free of contradictions and that they are in fixed relationships to proven concepts that have already been previously established. ... Once a number satisfies all these conditions, it can and must be considered as existing and real in mathematics. ... On the other hand, every superfluous constriction of the impulse for mathematical research seems to me to bring with it a much greater danger ... for the essence of mathematics lies precisely in its freedom. (Cantor 1883a, § 8)*

The underlying disagreement concerns what entitles us to admit a mathematical object into our ontology in the first place. For Cantor, the development of mathematics can itself justify an expansion of what exists mathematically: objects introduced through new theoretical constructions acquire legitimacy through the demonstrable relationships they bear to existing mathematics. This makes mathematical ontology responsive to mathematical practice, insofar as the development of new theories can motivate commitments to objects that were not previously recognized. Kronecker's position places greater weight on the arithmetic (and ultimately, intuitive – construct a number system using your fingers!) foundations from which such constructions must proceed, thereby restricting the extent to which mathematical practice can determine its own ontology.

It would be glib of me, but not entirely inaccurate, to describe the subsequent episodes in the foundational crisis as footnotes to the attitudes set by Kronecker and Cantor. Consider, for example, the work of Gottlob Frege. On the one hand, Frege (at the time of *Die Grundlagen der Arithmetik*) appeared to hold Cantor's basic conceptual breakthrough in high regard:

> *I find special reason to welcome in Cantor's investigations an extension of the frontiers of science, because they have led to the construction of a purely arithmetical route to higher transfinite numbers (powers). (Frege 1884; Frege 1950, § 86)*

However, Frege's own project was to establish a rigorous logical foundation for arithmetic, and this led him to question some of the prevailing accounts of what numbers are and how they are derived. In particular, he

rejected the idea that number could be explained simply by abstracting from the differences between objects and treating them as identical units (as Cantor sought to do). His objection was that this procedure seemed to produce a general concept under which objects fall, rather than an account of number itself:

> *And with that we are back once more at our arbitrary way of regarding things, unless we are willing, regardless of truth, to ascribe to things an identity going beyond any they actually possess. ... In that event we are not left, as Lipschitz maintains, with "the concept of the Number of the things considered"; what we get is rather a general concept under which the things in question fall. (Frege 1884; Frege 1950, § 34)*

Frege's disagreement with David Hilbert brought these questions into particularly sharp focus, culminating in a controversy over the foundations of geometry and the role of consistency proofs in mathematical justification *(Frege 1980)*. In *Grundlagen der Geometrie (Hilbert 1899)*, Hilbert demonstrated how the consistency of geometrical axioms could be reduced to that of arithmetic by interpreting geometrical terms through arithmetic objects and relations. For Frege, axioms expressed thoughts about a determinate subject matter, and consistency alone could not establish the existence of the objects concerned. Hilbert's approach placed the emphasis on the formal relationships specified by axioms, with consistency providing grounds for accepting a mathematical theory. But in another way, his program carried forward Cantor's conception of mathematical freedom, developing a systematic method for establishing the legitimacy of mathematical theories through their axiomatic structure.

On Kronecker's side, the expression of similar finitist and constructivist inclinations after him perhaps came from Émile Borel:

> *One can adopt a more strictly arithmetical point of view, as Kronecker did in algebra ... This new point of view can be characterized by the fact that, in any given problem, one introduces only a finite number of integers by means of which all the elements of the problem are explicitly defined.*
>
> *...*
>
> *In this construction, the idea of magnitude plays no part at all ... given the procedure itself, we can define only a countable infinity of such numbers. Some minds will see this as a deficiency; others, on the contrary, will think that, in practice, every number, in order to be effectively known, must be capable of being characterized by a finite number of words ...*

A related position emerged among several French mathematicians around the turn of the 20th century, including Émile Borel, René Baire, and Henri Lebesgue *(Hadamard 1905)*, who were later grouped under the label of *pre-intuitionists* in the terminology of the later Luitzen Jan Egbertus Brouwer *(Brouwer 1952)*.

Borel's approach placed greater restrictions on mathematical existence than Cantor's: consistency alone was insufficient, since mathematical objects also had to be effectively definable, in principle through a finite number of words. Given such a restriction, only a countable infinity of numbers could be individually characterized, placing substantial limits on the legitimacy of arguments involving the uncountable infinite. These mathematicians did not constitute a unified philosophical school, and their views differed in important respects. Nevertheless, their work raised a question that Brouwer would pursue much further: whether the restrictions required for mathematical knowledge should extend beyond the objects we admit to the very logical principles through which we reason about them.

Brouwer's position took the preceding disagreements in a rather different direction by questioning the assumption that mathematical objects could be understood independently of the constructions through which they become known. For Brouwer, mathematics is fundamentally a creation of the mind, preceding both mathematical language and the logical principles used to describe it. Its foundation lies in the intuition of time: the experience of one moment giving way to another, while the former is retained in memory. Abstracting from the particular content of this experience produces what Brouwer called the intuition of *two-ity (Brouwer 1952)*, from which the natural numbers arise. On this view, logic describes regularities within mathematical activity, and cannot independently establish the existence of mathematical objects.

Brouwer's conception of the real numbers (a mathematical structure that has been the subject of much controversy) makes the consequences of this position particularly clear. In his later intuitionistic mathematics, real numbers could be represented by infinite sequences of rational approximations, including what he called *choice sequences (Brouwer 1918)*. Some of these sequences are governed by a fixed rule, while others may develop through successive free choices whose outcomes are not determined in advance. This provided Brouwer with a conception of the continuum that was not restricted to numbers individually characterizable through finite descriptions, as in the approaches associated with Borel. At the same time, the infinite sequences involved were understood through their ongoing construction, rather than as completed infinite totalities. A real number could therefore have mathematical significance even when its successive approximations were not all determined by a single law.

On Brouwer's view, since a mathematical statement is established through a construction that proves it, he rejected the unrestricted use of the law of excluded middle, according to which every proposition P satisfies $P \vee \neg P$. For certain statements concerning real numbers, neither the statement nor its negation need be presently constructible. For example, one cannot generally assert that every real number is either equal to zero or unequal to zero without a method that establishes one of these alternatives. His approach thus connected the epistemology of mathematics directly to its metaphysical commitments: the nature of

mathematical objects depended on the constructions through which they could be brought into existence. As expressed in his doctoral thesis *(Brouwer 1907; Brouwer 1975)*:

> *We start with the notion of the natural numbers, etc. They are so familiar to us, that it is difficult to reduce this notion to simpler ones. Yet I shall try to describe their sense in plain words. In the perception of an object we conceive the notion of an entity by a process of abstracting from the particular qualities of the object. We also recognize the possibility of an indefinite repetition of the conception of entities. In these notions lies the source of the concept of natural numbers.*

Brouwer's position raises a puzzle: if mathematical truth is grounded in mental constructions, what becomes of the logical principles traditionally used to establish mathematical truths? This question motivated the work of Arend Heyting, a student of Brouwer who sought to give a formal account of intuitionistic mathematics. In 1930, Heyting *(Heyting 1930a; Heyting 1930b; Heyting 1930c)* published a systematic formalization of a logic for intuitionism (something Brouwer himself would likely have been skeptical of), including propositional and predicate logic, alongside formal systems for arithmetic and analysis. His contribution was to articulate the logical consequences of Brouwer's philosophical position, showing how mathematical reasoning could be governed by precise rules that reflected the constructive character of mathematical knowledge. In particular, the law of excluded middle could no longer be assumed indiscriminately, since establishing a disjunction requires a construction that establishes one of its alternatives.

Heyting's work thus introduced a distinction between the formal structure of a proof and the constructive meaning that its logical rules were intended to capture. An intuitionistic proof of an implication, for example, requires a construction that transforms a proof of its antecedent into a proof of its consequent; a proof of a disjunction requires a proof of at least one of its disjuncts. These interpretations connect the validity of logical inferences to the constructions that justify their conclusions. In this sense, Heyting provided a framework within which Brouwer's epistemological commitments could be investigated through formal logic, even though Brouwer himself regarded mathematical intuition as prior to its formal representation.

Gerhard Gentzen's *(Gentzen 1935a; Gentzen 1935b)* subsequent work brought these concerns into the development of modern proof theory. Through his systems of natural deduction and sequent calculus, Gentzen provided formal frameworks for investigating the structure of mathematical proofs, including both classical and intuitionistic reasoning. His work on the elimination of cuts, in particular, demonstrated that formal derivations possess structural properties that can be studied independently of the particular mathematical claims being proved.

1.3 Natural deduction, cut elimination, and Hilbert's paradise

Gentzen's natural deduction systems were developed to formalize mathematical reasoning through assumptions and their discharge. To prove an implication $A \to B$, for example, one assumes A, derives B, and then discharges the assumption:

$$\begin{array}{c} [A]^i \\ \vdots \\ B \end{array} \quad \frac{}{A \to B}\ (\to I, i) \qquad \frac{A \to B \qquad A}{B}\ (\to E).$$

The introduction rule specifies how an implication can be established, while the elimination rule specifies how it can be used. Analogous pairs of rules govern conjunction, disjunction, and the quantifiers. In this way, the structure of a mathematical proof can be studied through the rules by which propositions are introduced and used.

However, Gentzen's investigation of proof normalization led him to develop a second formalism, the sequent calculus, which provided a particularly powerful framework for analyzing derivations in both classical and intuitionistic logic. A sequent has the form

$$\Gamma \vdash A,$$

where Γ is a collection of assumptions and A is the conclusion. The notation makes explicit the relation between assumptions and conclusions that natural deduction tracks through its subproofs. More importantly for our purposes, it permits the structure of a proof to be investigated independently of the particular mathematical content of its formulas. Consider the following rules for conjunction and implication:

$$\frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \wedge B}\ (\wedge R) \qquad \frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C}\ (\wedge L)$$

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \to B}\ (\to R) \qquad \frac{\Gamma \vdash A \qquad \Gamma, B \vdash C}{\Gamma, A \to B \vdash C}\ (\to L).$$

Here, the left and right rules describe how the same logical operation functions when a formula occurs among the assumptions or as the conclusion. The remaining rules have analogous forms. For intuitionistic logic, the right side of a sequent contains at most one formula; the classical sequent calculus permits multiple formulas on the right. This distinction is important because the choice of logical system determines which mathematical inferences are admissible. In particular, the intuitionistic calculus does not admit the unrestricted law

of excluded middle as a theorem.

These positions therefore motivate two questions that proof theory allows us to investigate separately. What can be derived from a given mathematical system (a la Cantor, via Hilbert), and what restrictions should be placed on the means through which a mathematical conclusion is established (a la Kronecker, via Brouwer)? The classical and intuitionistic calculi provide different answers to the second question while permitting a common mathematical investigation of the first. The cut elimination theorem is especially significant in this respect, since it establishes a fundamental structural property of derivations in both systems.

The rule that the theorem eliminates is known as *cut*:

$$\frac{\Gamma \vdash A \qquad \Delta, A \vdash C}{\Gamma, \Delta \vdash C}\ (\text{Cut}).$$

This rule permits a proposition established in one derivation to be used as an assumption in another. The proposition A disappears from the conclusion, while the assumptions required to establish it are incorporated into the resulting derivation. The rule is useful for constructing proofs, since intermediate lemmas can shorten and organize a derivation. Gentzen's Hauptsatz establishes that every derivation in the standard sequent calculi for classical and intuitionistic logic can be transformed into one that contains no applications of cut.

We prove the intuitionistic version *(Troelstra and Schwichtenberg 2000)*. Let LJ denote a single-succedent sequent calculus for first-order logic, with formulas generated from atomic formulas and $\bot$ using $\wedge, \vee, \to, \forall, \exists$. Negation is understood as an abbreviation for $A \to \bot$. Contexts are finite sets of formulas, so exchange and contraction are implicit. Weakening is admissible. We use the standard context-sharing presentation of the logical rules; these rules are equivalent to the usual formulations in the presence of the structural properties of the calculus.

The initial sequents and the rule for falsity are

$$\frac{}{\Gamma, A \vdash A}\ (\text{Id}) \qquad \frac{}{\Gamma, \bot \vdash C}\ (\bot L).$$

The propositional rules are

$$\frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C}\ (\wedge L) \qquad \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \wedge B}\ (\wedge R)$$

$$\frac{\Gamma, A \vdash C \qquad \Gamma, B \vdash C}{\Gamma, A \vee B \vdash C}\ (\vee L)$$

$$\frac{\Gamma \vdash A}{\Gamma \vdash A \vee B}\ (\vee R_1) \qquad \frac{\Gamma \vdash B}{\Gamma \vdash A \vee B}\ (\vee R_2)$$

$$\frac{\Gamma \vdash A \qquad \Gamma, B \vdash C}{\Gamma, A \to B \vdash C}\ (\to L) \qquad \frac{\Gamma, A \vdash B}{\Gamma \vdash A \to B}\ (\to R).$$

The quantifier rules are

$$\frac{\Gamma, A[t/x] \vdash C}{\Gamma, \forall x\, A \vdash C}\ (\forall L) \qquad \frac{\Gamma \vdash A[y/x]}{\Gamma \vdash \forall x\, A}\ (\forall R)$$

$$\frac{\Gamma \vdash A[t/x]}{\Gamma \vdash \exists x\, A}\ (\exists R) \qquad \frac{\Gamma, A[y/x] \vdash C}{\Gamma, \exists x\, A \vdash C}\ (\exists L).$$

Here, $A[t/x]$ denotes capture-avoiding substitution of the term t for the free occurrences of x in A. In $\forall R$, the eigenvariable y must not occur free in Γ; in $\exists L$, it must not occur free in either Γ or C. Eigenvariables may always be renamed to satisfy these conditions.

Theorem. *If $\Gamma \vdash C$ is derivable in* LJ*, then it has a derivation containing no applications of* Cut.

Proof. Write $h(\pi)$ for the height of a derivation π, and define the rank of a formula by

$$\begin{aligned} r(P) = r(\bot) &= 0, \\ r(A \circ B) &= 1 + \max\{r(A), r(B)\}, \quad \circ \in \{\wedge, \vee, \to\}, \\ r(Qx\, A) &= 1 + r(A), \quad Q \in \{\forall, \exists\}. \end{aligned}$$

We first prove that cut is admissible for cut-free derivations. Suppose that π is a cut-free derivation of $\Gamma \vdash A$ and ρ is a cut-free derivation of $\Delta, A \vdash C$. We prove that $\Gamma, \Delta \vdash C$ has a cut-free derivation by lexicographic induction on

$$\big(r(A),\, h(\pi) + h(\rho)\big).$$

If π ends in an initial sequent, then its context already contains A, and the desired conclusion follows from ρ by admissible weakening. If ρ ends in an initial sequent, either C already occurs in Δ, in which case the

conclusion is initial, or $C = A$, in which case the conclusion follows from π by weakening.

Suppose next that the cut formula is not principal in the last inference of either derivation. The cut can then be permuted above that inference. If the inference has multiple premises, we apply the induction hypothesis to each premise containing the relevant occurrence and then reapply the inference. Each new cut has the same rank and a strictly smaller sum of derivation heights. We may therefore assume that both final inferences are principal on A.

Let's now consider propositional cases ($A = B \wedge D$, $A = B \vee D$, or $A = B \to D$, proofs in the quantificational case are similar). The principal cut is replaced by one or two cuts on the proper subformulas B and D. For example, if $A = B \to D$, we first cut $\Delta \vdash B$ and $\Gamma, B \vdash D$ to obtain $\Gamma, \Delta \vdash D$, and then cut this with $\Delta, D \vdash C$ to obtain $\Gamma, \Delta \vdash C$. For conjunction and disjunction, we similarly cut on the relevant subformulas introduced in the premises. In all such instances, the new cut formulas have strictly smaller rank than A, so the induction hypothesis applies to eliminate the new cuts.

Since the rank of $B[t/x]$ is strictly smaller than that of the quantified formula, the induction hypothesis applies to eliminate the cut. Atomic formulas and $\bot$ have no right-introduction rules. They therefore cannot form an additional principal case. All possibilities have been exhausted, establishing cut admissibility for cut-free derivations. Finally, consider an arbitrary derivation, possibly containing cuts. Proceed by induction on its height, replacing each immediate subderivation by a cut-free derivation. If its last inference is logical, reapply that inference to the cut-free premises. If its last inference is a cut, apply the cut-admissibility result just proved. The result is a cut-free derivation of the same end sequent. Hence every derivable sequent in LJ has a cut-free derivation. □

The significance of the theorem is that cut-free derivations possess a strong structural property. Every formula occurring in such a derivation is a subformula of a formula in the end sequent, up to the substitution instances introduced by quantifier rules. Intermediate lemmas may considerably shorten a proof, but the cut-elimination theorem shows that they are unnecessary for establishing derivability in the calculus. Their use can be eliminated through transformations of the derivation itself.

The corresponding result for natural deduction (more elusive) is the normalization theorem. An introduction rule followed immediately by its corresponding elimination rule constitutes a detour: for example, one may derive $A \wedge B$ from proofs of A and B, and then immediately extract A. Normalization replaces this sequence with the original proof of A. Cut elimination and normalization are closely related, although they are theorems concerning different proof formalisms. Gentzen's sequent calculus method was particularly important because it gave a uniform approach to the classical and intuitionistic cases.

As fruits of the long march of the foundational crisis, these results give us a mathematical vocabulary for studying the internal structure of proofs. Until recently, many questions about that structure were difficult to pursue systematically: proofs were produced to establish particular results, and their internal organization was usually examined in the context of individual arguments. AI-generated formal mathematics changes the scale of this inquiry. We now have large collections of machine-written derivations whose validity can be checked and whose structure can be examined directly.

If the proof assistant settles the question of formal correctness relative to its environment, do machine-generated derivations exhibit distinctive structural regularities? How do they compare with those produced by human mathematicians? And can the tools of proof theory help us answer these questions empirically? The rest of this paper suggests that, within limits, they can.

2 An empirical approach to proof theory

2.1 Proofs as programs

Before we can measure proofs, we need to say what a formal proof is in practice. A *proof assistant* is a computer program that checks mathematical proofs. The user states a theorem in a precise formal language and writes a proof of it. The program then checks every step of that proof against a small set of logical rules. If any step does not follow from the rules, the proof is rejected. The proof assistant we will use to study machine-generated proofs is Lean *(de Moura and Ullrich 2021)*.

Lean rests on an idea known as the Curry–Howard correspondence, or *propositions as types (Curry and Feys 1958; Howard 1980)*. Recall the natural deduction rules for implication from the previous section. To prove $A \to B$, we assume A, derive B, and discharge the assumption. To use $A \to B$, we combine it with a proof of A and obtain B. Now consider a function in a programming language. A function of type $A \to B$ takes an input of type A and returns an output of type B. Writing such a function means assuming a variable h of type A and building something of type B from it. Applying the function to an input of type A gives an output of type B:

$$\frac{\begin{array}{c}[h : A] \\ \vdots \\ q : B\end{array}}{\lambda h.\, q \;:\; A \to B}\,(\to I) \qquad \frac{f : A \to B \qquad p : A}{f\, p \;:\; B}\,(\to E).$$

Here, $\lambda h.\, q$ is the function that takes h and returns q, and $f\, p$ is the function f applied to p. A proposition is a type, and a proof of the proposition is a program of that type. Checking a proof is the same as checking

that a program has the type it claims to have. The same correspondence extends to conjunction, disjunction, the quantifiers, and much more. Lean uses a rich version of it known as dependent type theory *(Coquand and Huet 1988)*, in which types may depend on values (e.g. the type of vectors of length n).

In practice, people rarely write these programs by hand. Instead they write *tactics*, which are commands that build the program step by step. Consider the following small theorem:

```
theorem two_steps (a b c : Nat) (h1 : a <= b) (h2 : b <= c) : a <= c + 1 := by
  have h3 : a <= c := Nat.le_trans h1 h2
  exact Nat.le_succ_of_le h3
```

The theorem says that if $a \leq b$ and $b \leq c$, then $a \leq c + 1$. The proof first establishes the intermediate claim $h_3 : a \leq c$ using transitivity. It then concludes using the fact that anything at most c is at most $c + 1$. Lean reads these tactics and produces a single program, called the *proof term*. This process is called *elaboration*. A small trusted part of Lean, the kernel, then checks the proof term. The kernel never sees the tactics.

So, the text of a Lean file is what a person (or a machine) typed. The proof term is the derivation that the text describes. To get the proof term, the file has to be compiled. It also has to be compiled against the exact version of the library it was written for, since names and lemmas change between versions.

The library we will compile against when using Lean is mathlib *(The mathlib Community 2020)*. Mathlib is a single large library of formalized mathematics, written by hundreds of human contributors over several years. It covers algebra, analysis, topology, number theory, and much else. Its proofs are reviewed by other contributors before they are accepted, and it has its own conventions for how proofs should be written. At least until recently, almost all serious work in Lean, human or machine, was built on top of it–in some sense, it constitutes the "gold standard" for Lean proofs (though of course, in some sense, it is also an arbitrary choice, as we will see).

2.2 The data and language

As discussed, our human baseline is mathlib, at the version used by OpenAI (commit `d13f23b`) *(The mathlib Community 2026)*. The main AI-written corpus of mathematics we will use for comparison is OpenAI's `openai/math` library *(OpenAI 2026b)*. This repository contains 524 projects, and we compiled 96 of them, chosen at random (which was about as large a sample I could run on my personal MacBook Air). These samples contain about 36,000 theorems. I also compiled three smaller AI-written corpora from other systems. The first is a formalization of sphere packing by Math Inc's Gauss *(Math, Inc. et al. 2026; Hariharan et al. 2026)*. The second is a set of solutions to the 2025 Putnam competition by AxiomProver *(Axiom*

Math 2026a). The third is a set of solutions to the 2025 IMO by Aristotle *(Harmonic 2025; Achim et al. 2025)*.

Of course, it must be acknowledged (as we began to above) that mathlib is a polished library, and it is not the only kind of human Lean. We therefore added a second human reference, made of research projects (the Carleson project *(van Doorn et al. 2026; Becker et al. 2024)* and the formalization of sphere eversion *(Massot et al. 2026)*). These were written by humans for a particular result, much like the AI-written projects (though, compared to the text of mathlib, we suspect these libraries are very scarce). To control the quality of the data, the unit of every group comparison is the project. Theorems in the same project share authors and habits, so they are not independent observations. I also only used projects with at least 20 theorems. When I measure single proofs, I only use proofs of at least 10 steps. Imposing these standards, there are about 143,000 proofs in mathlib and 26,000 in OpenAI's library.

The simplest comparison between human (as in, mathlib-written) and AI-generated proofs looks only at their text. A language model assigns a probability to each next word (or *token*) of a text, given the words before it. If the model finds a text surprising, the text "costs" many bits per token. I call this cost, in bits per token, the *energy* of the text. We trained a simple language model *(Hindle et al. 2012)* (a 5-gram model, which looks at the previous four tokens) on mathlib's source code. We then asked how surprising it found OpenAI's source code.

OpenAI's text costs 7.00 bits per token, and mathlib's own (held-out) text costs 5.15. A sample of OpenAI's text costs more than a sample of mathlib's 94% of the time. This probability is called the *AUC (Hanley and McNeil 1982)*: the chance that a randomly chosen member of one group scores higher than a randomly chosen member of the other, so that 0.5 means no difference and 1 means complete separation. But 76% of the gap comes from identifiers. These are the names of lemmas, hypotheses, and definitions. Two proofs can use the same steps and different names. They can also use different steps and the same names. The text therefore mostly measures how proofs are named. To measure how they work, we need the proof terms–which, as discussed previously, are programs.

One might suspect that OpenAI simply writes about different mathematics than mathlib does. This does not explain the gap. Mathlib and OpenAI's library share 17 areas of mathematics with at least three projects each, and in every one of them OpenAI's median project costs more. Against projects in its own area, the typical OpenAI project pays 1.53 extra bits per token. The human research code sits in between (Figure 1), at 6.67 bits per token (in the text study, it also includes the FLT and PFR projects *(Buzzard and Taylor 2025; Anderson et al. 2023)*). Its gap to OpenAI is much smaller (0.33 bits per token, with an AUC of 0.74).

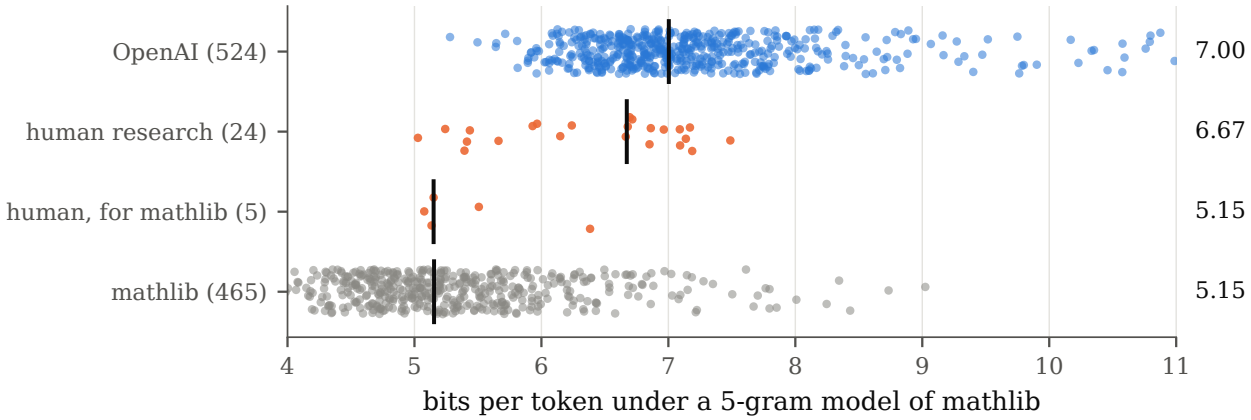


Figure 1: Text energy of each project (one dot per project with at least 20 theorems) under a 5-gram model of mathlib's source. Each mathlib project is scored by a model trained on the rest of mathlib. Ticks mark medians; values above 11 bits are drawn at the edge. The row "human, for mathlib" holds the directories of the human research projects that were written to be added to mathlib.

Interestingly, the parts of the human research projects that were written to be added to mathlib later cost what mathlib costs (5.15, over two projects). So the language model measures how closely a text follows mathlib's conventions.

2.3 What is already measured

Most of the quantities in this paper are not new. They have been computed, often at scale, by the people who build automated provers and the tools around proof assistants, where they serve as objectives to improve. The first language-model prover for Lean already found that more than a third of its proofs were shorter, and produced smaller proof terms, than mathlib's own *(Han et al. 2021)*. Recent provers *(Hubert et al. 2025; Ren et al. 2025; Wang et al. 2025; Lin et al. 2025; Chen et al. 2025)* often write proofs that are correct but long, and a line of work now shortens them. ImProver rewrites proofs to improve a chosen metric such as their length *(Ahuja et al. 2024)*. ProofOptimizer trains a model to simplify proofs, and more than halves their length on competition benchmarks *(Gu et al. 2025)*. LeanPolish releases tens of thousands of verified local edits that compress Lean proofs *(Bourigault 2026)*. Others refactor generated proofs into reusable lemmas *(Fu et al. 2026)* or evaluate their quality with tools *(Caldwell 2026)*, and mathlib's own linters flag tactics that change nothing *(The mathlib Community 2026)*. Kallel and El Louadi measure how much kernel-checked proof stands behind the statements of a release of AI-written results *(Kallel and El Louadi 2026)*. Older work on interactive theorem provers measured proof scripts *(Wiedijk 2000; Aspinall and Kaliszyk 2016)*, refactored them *(Whiteside et al. 2011)*, and mined or compressed them by introducing lemmas and definitions *(Kaliszyk and Urban 2013; Blanchette et al. 2015; Vyskočil et al. 2010)*. The work on cut introduction compresses proofs by finding the cuts they lack *(Hetzl et al. 2014a; Ebner et al. 2018)*,

and the realized compression below is a crude relative of it.

What these share is an aim: a shorter proof, a faster check, a better prover. My debt to these prior pieces of work is large because they shaped a core intuition I had guiding the work in this paper–that AI generated proofs are somehow large in an "unnatural" way (at least for humans), and that it was possible to measure the size of a proof term or the number of its wasted steps as formalized in Lean in a rigorous way. What I try to add is a way of using such quantities as *measurements* of proofs rather than as targets to optimize them against. That means comparing them against how much a reference library differs from itself, or against the proof's own cut-free form; checking them against theorems they must obey; and controlling for how the proofs were compiled and for the tactics that wrote them. The hope is that, made reliable in this way, they can eventually be used to study properties of formal proofs proofs themselves at scale.

2.4 Preparing the corpora

Every number in this section depends on how the corpora were read, cleaned, and compiled, so I describe this in some detail. All of the code for this study (the language model, the Lean programs that extract proof terms, and the statistics) was written by Claude Code *(Anthropic 2026)*, Anthropic's coding agent, working at my direction. I chose the questions, the measures, and the controls. The code, the job files, the regression tests, and the result files behind every table and figure are available in the accompanying repository *(Aryal 2026)*, with instructions for rerunning them. This way of working has one particular risk: code that is cheap to write is also cheap to change, and a small change in how a corpus is cleaned can move a result. So the controls had to be maintained deliberately. Every corpus went through the same pipeline, against the same mathlib, with the same thresholds. When a part of the pipeline changed, every corpus was rerun through it. Where a result moved because of such a change, I report the change where the result appears.

The text study works directly on the source files, without running Lean, so it covers all 524 OpenAI projects. The text was prepared as follows:

1. A lexer (a program that splits source text into tokens) strips comments and string literals, and splits each file into declarations (theorems, lemmas, and definitions).

2. Identifiers are split into the pieces mathlib uses to build its names. For example, `Nat.le_succ_of_le` becomes `Nat`, `le`, `succ`, `of`, `le`. A new name built from familiar pieces is then cheap, and only genuinely new vocabulary is expensive.

3. A piece the model has never seen is coded as an "unknown" token followed by its spelling, character by character. So the energy of a text is a proper code length: the number of bits needed to transmit

it.

4. Numerals of four or more digits are coded digit by digit and booked separately. Otherwise, a few proofs that carry long numerical certificates would dominate the cost of their whole project.

5. At most 1,000 declarations are sampled from each non-mathlib project. All of mathlib is used.

6. Mathlib's projects are divided into five folds. Each mathlib project is scored by a model trained on the other four folds. Without this, mathlib would be scored by a model that had already seen it, and its energy would be artificially low.

7. The model is an interpolated Kneser–Ney model *(Kneser and Ney 1995; Chen and Goodman 1999)*, a standard smoothed n-gram model. Its order (five) was chosen among orders three to six by its cross-entropy on held-out mathlib text.

Each token is also classed as a name piece, a numeral, or an operator. This is how the gap was split above. Of the pooled gap of 1.83 bits per token, 76% is carried by name pieces, and syntax adds only 0.28 bits.

The compiled study needs much more care, since a proof term exists only once Lean has elaborated the file. The pipeline is as follows:

1. All corpora are compiled with Lean 4.34.1 against mathlib at commit `d13f23b`, which I built on my machine. OpenAI's full library has 26 million lines, which is too large to compile with 8 GB of memory, hence the sample.

2. A Lean program (the *driver*) elaborates each file and walks every proof term. For each theorem it writes out the gestures, the cut records, and the proof-term measures described below.

3. A *module* is a single Lean file, and its *environment* is the set of definitions and notation it can see. The OpenAI projects were compiled in one shared environment, one module after another. This is faster than Lean's normal build, which gives each module an environment made only of its own imports. In a shared environment, notation declared in one module can leak into the next. The human projects were compiled module by module, each with its own imports, as Lean's build tool would. To check whether the compilation mode matters, I compiled nine OpenAI projects both ways. The gestures were identical for 97.2% of 4,089 theorems, and the divergence moved from 280.4 to 280.5 bits.

4. Several engineering failures had to be handled without biasing the sample. Lean never frees memory for repeated imports, so the module-by-module driver restarts every eight modules. A module that stalls is skipped after a time limit and recorded; one of the 97 drawn OpenAI projects stalled for more

than 45 minutes and is excluded. An early version of the driver also used a Lean interface that silently drops error messages, which could have hidden failing theorems. It was replaced.

5. Theorems containing `sorry` (Lean's placeholder for a missing proof) are dropped everywhere. Theorems that do not compile under this mathlib are dropped, at the rates given below.

6. Lean generates auxiliary theorems automatically, such as equation lemmas for definitions and the `_proof_n` lemmas it splits off long proofs. These are folded back into the declaration that produced them, so that a proof is not counted twice because the compiler cut it in two.

7. As discussed below, `omega` and `lia` are merged into one gesture.

I also checked a sample of extracted proofs by hand against their source.

At the end of this pipeline, each theorem is represented by its proof term, the program that Lean's kernel actually checks. The question is then what to measure on these terms, given that the corpora prove different theorems.

2.5 Gestures

A proof term is a tree. Each node is one step of the proof, such as the use of a lemma, the application of a function, or the introduction of an assumption. The children of a node are the subproofs it depends on. Most of the detail at each node is specific to the theorem being proved. But, we want to compare proofs of different theorems, so we must find a way to strip this detail away.

Definition. *We assume a proof term is a tree. Then, define a gesture as a node of a proof term with three changes made to it. One, for a gesture, all names of hypotheses are erased. Then, every type is cut down to its head connective (e.g. an equation becomes $=$, an implication becomes $\to$). Finally, every hypothesis is labelled by how often it is used:* 0 *(never),* 1 *(once), or* ω *(more than once).*

For example, the transitivity of equality takes a proof of $a = b$ and a proof of $b = c$ and returns a proof of $a = c$. Every use of it becomes the same gesture,

$$\texttt{Eq.trans}(=, =) \Rightarrow =,$$

whatever a, b, and c are.

Erasing names and keeping only the shape of each step is Hilbert's attitude toward geometry, applied to proofs. For Hilbert, it did not matter whether the axioms spoke of points and lines or of tables and chairs *(Blumenthal 1935)*; what mattered were the relations the axioms laid down. A gesture likewise keeps the

relations between the steps of a proof and throws away what the steps are about. Of course, Frege's objection applies *(Frege 1980)* here too. Two proofs with the same gestures may be about entirely different things. We accept this loss because the corpora prove different theorems, so the shape of their steps is the only thing that can be compared across them.

We walk every proof term in full and record the gesture at each node. Each project then has a *gesture mix*, the proportion of its steps taken up by each gesture. To compare corpora, I kept the 300 most common gestures and gave every project equal weight (so that a few large projects cannot dominate). We then compare the spread of OpenAI's projects with the spread of mathlib's using the Kullback–Leibler divergence *(Kullback and Leibler 1951)*, which measures in bits how different two distributions are. If each corpus's gesture mixes are approximated by a normal (Gaussian) distribution, this divergence splits exactly into a sum over a set of directions. Each direction is a particular combination of gestures in which OpenAI's projects vary differently from mathlib's. I call these directions *eigenproofs*. The bits carried by the eigenproofs add up to the total divergence.

Any two sets of projects differ somewhat, and a sample of 96 projects differs from its population by chance alone. So the divergence has to be compared with how much mathlib differs from itself. To measure this, we draw random sets of 96 mathlib projects, and measure their divergence from the rest of mathlib. This is the *null*.

OpenAI's gesture mix is 280 bits from mathlib's. Random sets of mathlib projects of the same size are 49 bits away (and at most 58 bits, in ten draws). OpenAI's library is therefore 5.8 times as far from mathlib as the null. Human research code is 1.8 times as far from mathlib as its own null. About a third of OpenAI's divergence, and most of human research code's excess over its null, comes from the certificates that tactics write into proof terms (see "Is it only tooling?"). Figure 2 shows how the 280 bits are spread over the eigenproofs. Each of the first twelve exceeds the largest value the null produced at its rank. A theorem's *score* on an eigenproof is the projection of its gesture mix onto that direction. The table gives the share of theorems whose score lies more than two standard deviations (s.d.) above mathlib's mean. The first four eigenproofs are:

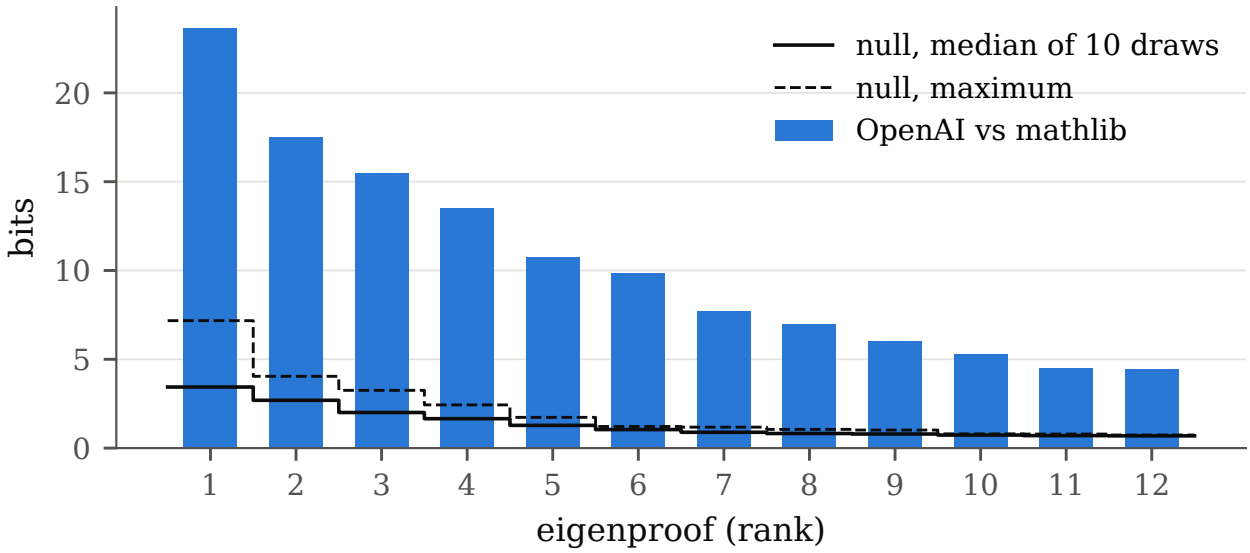


Figure 2: Bits carried by OpenAI's first twelve eigenproofs against mathlib, with the null: the same decomposition for random sets of 96 mathlib projects against the rest of mathlib (median and maximum over ten draws).

			theorems beyond +2 s.d.		
	eigenproof	bits	mathlib	OpenAI	human
1	integer arithmetic inside chains of equations	23.6	0.7%	6.3%	0.5%
2	rewriting under binders	17.5	0.2%	4.2%	4.4%
3	closing by evaluating numerals	15.5	0.6%	8.4%	11.2%
4	normalizing, then closing	13.5	0.9%	13.5%	12.8%

Only the first eigenproof belongs to OpenAI alone. It is the pattern of proving an equation by a call to an arithmetic decision procedure (Lean's `omega` *(Pugh 1991)*) in the middle of a chain of equations. The second rewrites inside a *binder*, a λ that introduces a variable. The others are habits that OpenAI shares with human research code, to a similar degree. When the data is split in half at random, the first direction is recovered on each half with a correlation of 0.95.

One choice affects these numbers. Mathlib recently switched from `omega` to a newer tactic, `lia`, which does the same job. OpenAI's library still uses `omega`. If the two are counted as different gestures, the divergence rises from 280 to 316 bits. I merged them, since the two tactics decide the same class of problems. The 36 bits between the two figures come entirely from a change in mathlib's tooling.

The gestures also let us count some simpler things. Each hypothesis is used never, once, or more than once, and each intermediate claim is a cut (we will return to cuts below). The medians over projects are:

	mathlib	OpenAI	human	AUC vs mathlib	AUC vs human
steps per theorem	18.5	27.0	29.0	0.75	0.40
unused hypotheses	8%	12%	14%	0.64	0.41
intermediate claims per 100 steps	0.36	1.20	0.64	0.86	0.79
unused claims	0.0%	0.9%	1.0%	0.74	0.50

Against mathlib, OpenAI's proofs are longer, state more claims, and leave more hypotheses unused. The claims and the unused hypotheses survive a control for automation, but the length mostly does not (see "Is it only tooling?"). Against human research code, only the extra claims stand out (AUC 0.79). Human research code is as long (AUC 0.40) and leaves as many hypotheses unused (AUC 0.41).

2.6 What mathlib confounds

So far, every comparison has been a comparison with mathlib. As discussed, mathlib is in some sense the gold standard for Lean proofs. But, nonetheless, it cannot be relied on its own to measure the *intrinsic* properties of mathematical proofs–in fact, at least five of its properties could each produce a difference between OpenAI and mathlib that has nothing to do with machines.

1. Mathlib has a *house style*. Its proofs are reviewed, shortened, and rewritten to follow its conventions before they are accepted. A proof that is not written the way mathlib writes proofs costs more under any model of mathlib, whether or not it is a worse proof. Energy therefore measures conformity to mathlib. The `omega` and `lia` example above is a small instance of this.

2. Mathlib has a different *purpose*: it is a library, and its lemmas are stated in general form so that others can use them. OpenAI's projects (like the Putnam and IMO solutions) are aimed at particular results. A research project leaves hypotheses unused and states claims it does not need, because it is unfinished. The human research code controls for this, at least partly. As we saw, unused hypotheses and dead claims are as common in Carleson and sphere eversion as in OpenAI's library. They may therefore be features of research formalization in general, rather than of machines. Of course, the human reference is small: 11 compiled projects from two efforts, against 1,193 mathlib projects. Its null is correspondingly large, and every interval around it is wide.

3. Mathlib is tied to a *version*. Every corpus has to be compiled against the same mathlib to be compared. The other AI corpora were written for older versions of Lean (4.20 to 4.28), and some of their theorems no longer compile. Sphere packing loses 20%, Putnam 11%, and IMO 67%. Even Carleson, written by humans, loses 222 of its 4,898 theorems. These failures are biased: longer proofs are more likely to

break, so the surviving samples lean towards shorter proofs.

4. A proof term records the *tools* as much as the author. When a proof calls `omega`, the tactic writes a *certificate* into the proof term (a proof that its answer is correct), and that certificate has its own gestures. Our first eigenproof is, in part, a measure of how often a tactic is called. I separate the two below ("Is it only tooling?"), by collapsing each certificate into a single step.

5. Finally, we must also consider the question of *subject matter*. Mathlib covers some areas much more heavily than OpenAI does. The text study controls for this by comparing within areas, as above. The gesture study does not, and it should be read with this in mind.

All five of these confounds share one structure. Each of them comes from measuring a proof against *other* proofs. A gesture mix is only unusual relative to some population, and the population I chose was mathlib. If I had chosen differently, the numbers would change. Brahe measured positions against the fixed stars, which are not truly fixed, and his data was still usable. But a measure that needs no reference population would remove most of these confounds at once (the tools, as we will see, remain). I now argue that cuts (and their eliminations, a la Gentzen) give us one.

2.7 Cuts in Lean

Recall the cut rule:

$$\frac{\Gamma \vdash A \qquad \Delta, A \vdash C}{\Gamma, \Delta \vdash C}\ (\text{Cut}).$$

A cut proves an intermediate claim A and then uses it as an assumption. This is exactly what the `have` tactic does in the example above. The proof first establishes $h_3 : a \le c$, and then uses h_3 to prove the theorem. Under propositions as types, the proof p of the claim and the proof q that uses it combine into the term

$$(\lambda h.\, q)\ p,$$

a function that takes h and returns q, applied to p. Lean has several ways of writing this (`have`, a `let` that names a proof, or a function applied directly to a proof). We count all of them as cuts.

Eliminating a cut means removing the intermediate claim. In terms of programs, we replace every occurrence of h in q by p itself, written $q[p/h]$. For the example above, the cut-free proof term is

$$\texttt{Nat.le_succ_of_le}\ (\texttt{Nat.le_trans}\ h_1\ h_2).$$

How much this costs depends on how often h is used, which is exactly what the labels 0, 1, and ω in our

gestures record. If h is never used, the cut is *dead*, and eliminating it simply deletes p. If h is used once, the cut is *linear*, and p moves to the place where it is used. Then, we say that if h is used many times, the cut is *shared*, and p is copied once for each use. It seems that only a shared cut can make a proof smaller.

2.8 Don't eliminate cut!

Gentzen showed that every cut can be eliminated. But, Statman and Orevkov *(Statman 1979a; Statman 1979b; Orevkov 1982)* later showed that the cost of doing so can be enormous. In fact, in the worst case, the cut-free proof is larger than any fixed tower of exponentials in the size of the original.

George Boolos drew a philosophical lesson from this in a short paper titled *Don't eliminate cut (Boolos 1984)*. He gave an inference in first-order logic that is valid, and that has a short derivation in a standard natural deduction system. The derivation fits on a few pages, and it works by proving a handful of lemmas and then using them. Boolos then showed that any derivation of the same inference in a standard cut-free system is unimaginably long. It could not be written down in the physical universe, let alone checked. His point was that cut elimination makes lemmas dispensable only in principle. The cut-free derivation is a finite object, so even Kronecker would grant that it exists. But in Boolos's example, only the derivation with lemmas can be written down and checked.

This gives us a way of looking at cuts that does not depend on mathlib. Every proof has a cut-free form, and we can ask of each cut whether it earns its place, by comparing the proof with its own cut-free form. A gesture mix is unusual only relative to a population. A cut is judged against the proof that contains it. In this sense, I would argue, cuts are more *intrinsic* to a proof than gestures are.

Of course, Boolos was concerned with the extreme case, in which a cut saves a tower of exponentials. We will see below that real libraries are nowhere near this case. Most cuts in mathlib are linear (74%), so they organize a proof without shortening it. The measures below ask, proof by proof, whether the author's cuts shorten it, and how much shorter it could have been.

2.9 Measuring cuts

The *size* of a proof term is the number of nodes in its tree. For each proof π, we compute three sizes. The first is the size of the proof as written, $|\pi|$. The second is the size of its cut-free form, $|\pi^{\mathrm{cf}}|$, with every cut eliminated. The third is the size of its optimal layout, $|\pi^{\mathrm{opt}}|$. This is the same proof with the best possible choice among the author's cuts. It keeps every cut that saves space and removes every cut that does not. (A cut used u times with a proof of size $|p|$ is kept when $(u-1)\cdot|p| > u+1$, that is, when the copies it saves

outweigh the cost of naming it.)

The cut-free form can be very large. Some proofs in mathlib consist of hundreds of nested cuts, and their cut-free forms are 2^{300} to 2^{450} times larger than the proofs as written. So we never build it. Instead we use *hash-consing (Filliâtre and Conchon 2006)*, which stores each distinct subterm only once. A subterm that appears a thousand times is stored once and counted a thousand times. This gives us the exact size, as a very large integer, without writing the proof out.

Definition. *The cut compression of a proof π is*

$$\kappa(\pi) = \log_2 \frac{|\pi^{\mathrm{cf}}|}{|\pi|}.$$

Its compressibility is

$$\chi(\pi) = \log_2 \frac{|\pi^{\mathrm{cf}}|}{|\pi^{\mathrm{opt}}|},$$

and its layout redundancy is

$$\rho(\pi) = \log_2 \frac{|\pi|}{|\pi^{\mathrm{opt}}|}.$$

All three are measured in bits. A proof that is half the size of its cut-free form has $\kappa = 1$. The optimal layout is never larger than the other two, so χ and ρ are never negative. The three are related by

$$\kappa = \chi - \rho.$$

The compressibility χ is how much a proof *could* save by its cuts. The redundancy ρ is how far the author's choice of cuts is from the best one. If $\rho > \chi$, then $\kappa < 0$, and the proof is larger with its cuts than without them. The `have` in our small example is a case of this. The claim h_3 is used once, so it saves nothing, and the extra step only adds to the size.

The quantity κ also has a direct reading. When we eliminate every cut, each step of the written proof is copied some number of times. One can show that κ is exactly $\log_2$ of the average number of copies of a written step. So one bit of cut compression means that, on average, each step would be written twice without the cuts.

2.10 Computing the cut measures

The definitions above leave several implementation choices open, and each one affects the numbers. I made them as follows:

1. *What counts as a cut.* A cut is a `have`, a `let` that binds a proof, or a function applied directly to a proof. Whether a term is a proof is decided from the declared type of its statement. A function applied to data (e.g. to a number) is substituted without being counted as a cut, because a proof cut can be hidden inside it.

2. *What counts as a use.* A use is an occurrence of the cut's variable in a proof. A proof that is mentioned only inside a type or a piece of data is counted as a single copy, since eliminating the cut never duplicates it. An early version counted every such mention as a copy, which inflated the use counts. This was found and fixed before any result was reported.

3. *Five normal forms.* A *normal form* here is a version of the proof with some class of cuts eliminated. For each proof, the driver computes the exact size of five versions: as written (N_0); with dead cuts removed (N_1); with linear cuts also removed (N_2); with every cut removed (N_3, the cut-free form); and in the optimal layout (N_4). The optimal layout is decided from the innermost cut outwards, keeping a cut used u times with a proof of size $|p|$ exactly when $(u-1)\cdot|p| > u+1$. The five sizes split κ into three parts,

$$\kappa = \underbrace{\log_2 \frac{N_3}{N_2}}_{\sigma} - \underbrace{\log_2 \frac{N_1}{N_2}}_{\ell} - \underbrace{\log_2 \frac{N_0}{N_1}}_{w},$$

   where σ is the gain from shared cuts, ℓ the cost of naming proofs that are used once, and w the cost of proofs that are never used.

4. *Exact sizes, averaged per proof.* Sizes are exact integers of arbitrary size. A corpus figure is the mean, over proofs, of each proof's bits. A ratio of summed sizes would be decided by a single mathlib proof whose cut-free form is 2^{450} times its written size. The first version of this analysis used such ratios, and was corrected before publication.

5. *Rates.* Rates such as "cuts per 100 steps" divide by the size of the proof with shared subterms counted once, so that proofs whose trees are enormous through repetition do not dilute every rate.

The realized compression of the next subsections needs a second, independent run of the driver over the proof terms, which I call the *hashing pass*. It eliminates every written cut by substitution, and stores the result only through hashes. Every subterm receives a *canonical hash*, a fingerprint computed from its structure, so that identical subterms receive identical hashes. A cut variable hashes as the proof it stands for, and an ordinary bound variable hashes as its binder. So the same subproof is a single node wherever it occurs, whether it was written out twice, named once and used twice, or substituted from a cut. Two occurrences of a subproof are identified only if their free variables (variables bound outside the subproof) are bound by

the same binders. This ensures that every node shared in the count could actually be named by a `have` at that point of the proof. The explicit layout G names a node when its number of references times its size exceeds its size plus its references plus one, and is then refined greedily up to six times. Each proof is given a budget of 200,000 distinct nodes, and proofs over budget are dropped (10 of 254,147 in mathlib). Logical rank unfolds definitions up to 24 times.

Before the hashing pass was run on any corpus, it was checked against a set of small theorems whose values were computed by hand. These remain in the repository as a regression test. The pass went through four versions. The first read ranks syntactically and failed the rank ladder; the second fixed the two errors behind this (described under "Checking the instruments"); the third also unfolds definitions when deciding whether a hypothesis is a proof; the fourth fixed the last exceptions to the ladder (also described there). Every corpus was run with the fourth version.

2.11 Cuts across corpora

Table values are means over proofs of at least 10 steps. Comparisons between projects use the projects with at least 20 such proofs (94 of OpenAI's 96). Figure 3 shows the same two quantities per project.

	mathlib	human	OpenAI	sphere packing	Putnam	IMO
proofs measured	143,448	3,195	25,928	3,553	353	29
proofs with a cut	17%	33%	47%	54%	73%	69%
cuts dead / linear / shared	1/74/25%	1/77/22%	2/82/16%	3/88/9%	4/86/10%	2/86/12%
χ (bits per proof)	0.029	0.063	0.035	0.022	0.056	0.056
ρ (bits per proof)	0.015	0.028	0.071	0.188	0.174	0.159
proofs larger with their cuts	13%	26%	40%	50%	67%	55%

AI-written proofs contain more cuts, and more of their cuts are linear. Comparing projects, OpenAI's redundancy ρ is higher than mathlib's with an AUC of 0.95 ($p = 2 \cdot 10^{-44}$; all p-values are from two-sided Mann–Whitney tests *(Mann and Whitney 1947)*), and higher than human research code's with an AUC of 0.85 ($p = 4 \cdot 10^{-5}$). Almost all of this redundancy comes from claims used exactly once. OpenAI's compressibility χ is only slightly higher than mathlib's (AUC 0.64, $p = 1 \cdot 10^{-5}$), and human research code is the most compressible of the three (OpenAI against human, AUC 0.22, $p = 9 \cdot 10^{-4}$). OpenAI's extra cuts therefore come from claims its proofs do not need. Even the slight excess in χ comes from automation (see "Is it only tooling?"). On proofs that call no automation, OpenAI is no more compressible than mathlib (AUC 0.55, $p = 0.14$), and its redundancy is as high as ever (AUC 0.94). Sphere packing and Putnam are more redundant still (AUC 0.95 and 1.00 against mathlib), and neither is clearly more compressible than

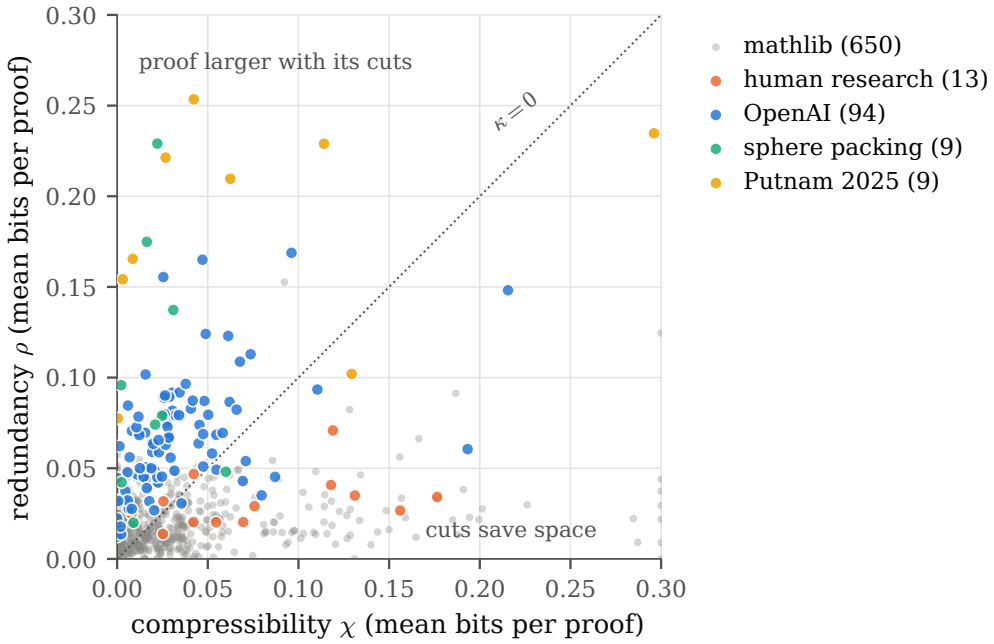


Figure 3: Mean compressibility χ and redundancy ρ per project (projects with at least 20 proofs). Above the diagonal, $\kappa < 0$ on average, and the project's proofs are larger with their cuts than without them. Values above 0.3 are drawn at the edge.

mathlib (AUC 0.58 and 0.70, $p = 0.42$ and 0.03). The IMO corpus is too small to test.

All of these values are small. Sharing could halve a proof (that is, $\chi \geq 1$) in at most 2.3% of proofs in any corpus. None of these corpora comes near the regime of Boolos's example.

I also measured cuts one level up, treating a project's own lemmas as cuts and their citations as uses. This result did not survive more data. With the first 67 OpenAI projects, OpenAI seemed to compress more at this level than mathlib. With all 94, the difference was no longer significant ($p = 0.11$). Inlining every local lemma multiplies the median project by about $2^{1.1}$ in mathlib and $2^{1.5}$ in OpenAI.

2.12 Is the divergence only layout?

OpenAI's redundant claims add steps, and every step has a gesture. So it is fair to ask whether the 280 bits of divergence come only from the redundant claims. To check, I reran the entire gesture analysis (vocabulary, eigenproofs, null, and tests) on three subsets. The first leaves out definitions, which have no cuts. The second keeps only proofs already in their optimal layout ($\rho = 0$). The third keeps only proofs with no cuts at all. These keep 30,157, 18,220, and 17,881 of OpenAI's 36,336 theorems.

The divergence survives (Figure 4). On proofs in their optimal layout it is still 4.6 times the null. The first eigenproof, however, is mostly layout: 76% of OpenAI's excess on it comes from proofs with redundant

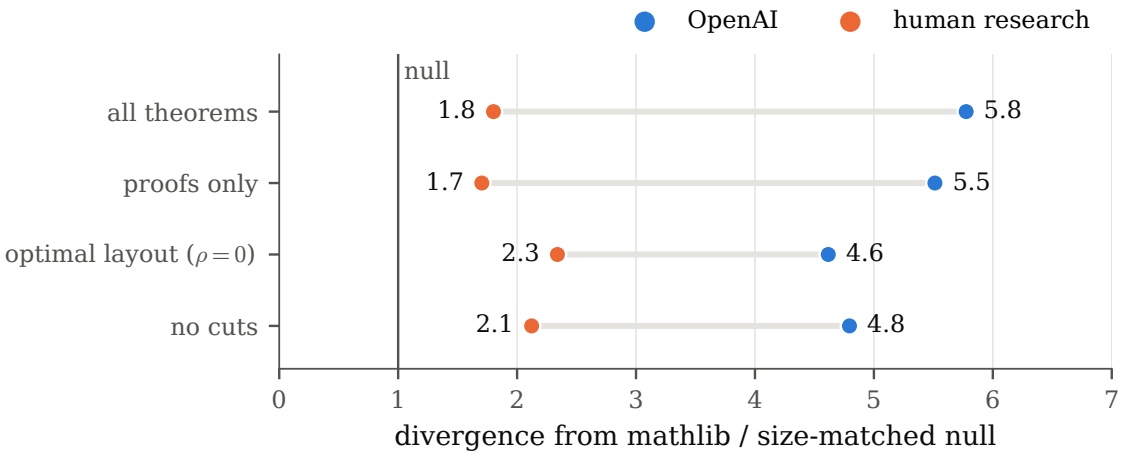


Figure 4: Divergence from mathlib as a multiple of the size-matched null, on all theorems and on the three layout-controlled subsets. Each subset reruns the whole pipeline (vocabulary, eigenproofs, null). OpenAI's divergence is 280, 299, 308, and 316 bits against nulls of 49, 54, 67, and 66.

layout. The typical case is a line like `have h : a = b := by omega`, a claim proved by arithmetic and used once. On proofs in their optimal layout, this direction drops to fifth place (13 bits), and the top direction becomes rewriting under binders (46 bits). Proofs with redundant layout carry 43% of OpenAI's excess on this direction, close to their 40% share of OpenAI's proofs. Even on these proofs, OpenAI scores beyond +2 s.d. on eigenproofs 2 to 8 between 13 and 31 times as often as mathlib does.

The same check changed one earlier result. A Kneser–Ney model trained on mathlib's proof terms predicts each gesture from its parent in the tree, and so assigns each proof a cost in bits per step. Over all theorems, this model found OpenAI's proofs no more surprising than mathlib's (3.27 bits per step for both, AUC 0.53). On proofs in their optimal layout, OpenAI costs more (3.37 against 3.13, AUC 0.63). The redundant claims are cheap and predictable steps, and they had been pulling OpenAI's average down. I had reported the tie before running this check. On the compiled sample, the text gap between median projects moves from 1.75 to 1.78 bits per token.

2.13 Realized compression

The redundancy ρ measures waste. But χ only measures the compression available among the author's own cuts. A proof may contain a repeated subproof that the author never named, and χ cannot see it. So we would also like to know how much compression a proof contains at all, whoever laid it out, and how much of it the author obtains.

For this, I eliminate every cut and then look for repeated subproofs in the result. Let S be the size of the cut-free proof. Then define three quantities, all in bits. One, $\chi_{\text{author}} = \log_2(S/A)$, where A is the size of the

author's own cuts in their optimal layout. This is what the author's cuts achieve. Two, $\chi_{\mathrm{lb}} = \log_2(S/G)$, where G is the size of an explicit layout that names any repeated subproof that pays for its own name, whether or not the author named it. Finally, $\chi_{\mathrm{DAG}} = \log_2(S/D)$, where D counts the distinct subproofs, that is, the nodes of the proof's *directed acyclic graph* (DAG), in which every repeated subproof is stored once. This is sharing with no cost for naming. If χ^* is the best achievable compression, then

$$\chi_{\mathrm{author}} \le \chi^* \qquad \text{and} \qquad \chi_{\mathrm{lb}} \le \chi^* \le \chi_{\mathrm{DAG}}.$$

The *realized share* of a corpus is

$$\frac{\sum_\pi \chi_{\mathrm{author}}(\pi)}{\sum_\pi \chi_{\mathrm{lb}}(\pi)}.$$

Since χ_{lb} is at most χ^*, this share is a conservative estimate. Importantly, S, D, and χ^* depend only on the cut-free proof. Any two layouts of the same proof (e.g. `have h := p; (h, h)` and `(p, p)`) give the same values.

	mathlib	human	OpenAI	sphere packing	Putnam	IMO
proofs measured	143,439	3,195	25,927	3,553	353	29
χ_{author}	0.029	0.063	0.035	0.022	0.056	0.056
χ_{lb}	0.039	0.105	0.115	0.162	0.186	0.173
χ_{DAG}	0.100	0.223	0.290	0.424	0.418	0.437
realized share	73%	61%	31%	14%	30%	33%
missed at least 1 bit	0.1%	0.3%	1.1%	3.2%	3.4%	3.4%

The hashing pass was a separate run, and it drops proofs over its budget, so its proof counts differ slightly from those in the first table. Mathlib realizes 73% of the compression its proofs contain, and human research code 61%. OpenAI realizes 31%, Putnam 30%, and sphere packing 14%. Sphere packing is the AI corpus whose gestures are closest to human code (1.4 times its null), and it realizes the least. Part of these gaps comes from automation (see "Is it only tooling?").

AI-written proofs also have more compression available (χ_{DAG} is two to four times mathlib's). This is mostly because they are longer and call more automation, whose certificates repeat themselves. On proofs that call no automation, OpenAI's χ_{DAG} is 1.5 times mathlib's. From 100 nodes up, proofs of the same size have about the same χ_{DAG} in every corpus, to within a factor of about 1.6. One might then object that longer proofs are simply harder to lay out well. To check this, I grouped proofs by the size of their cut-free form (Figure 5).

From 30 nodes up, mathlib realizes more in every bin. Across projects, OpenAI's realized share is lower

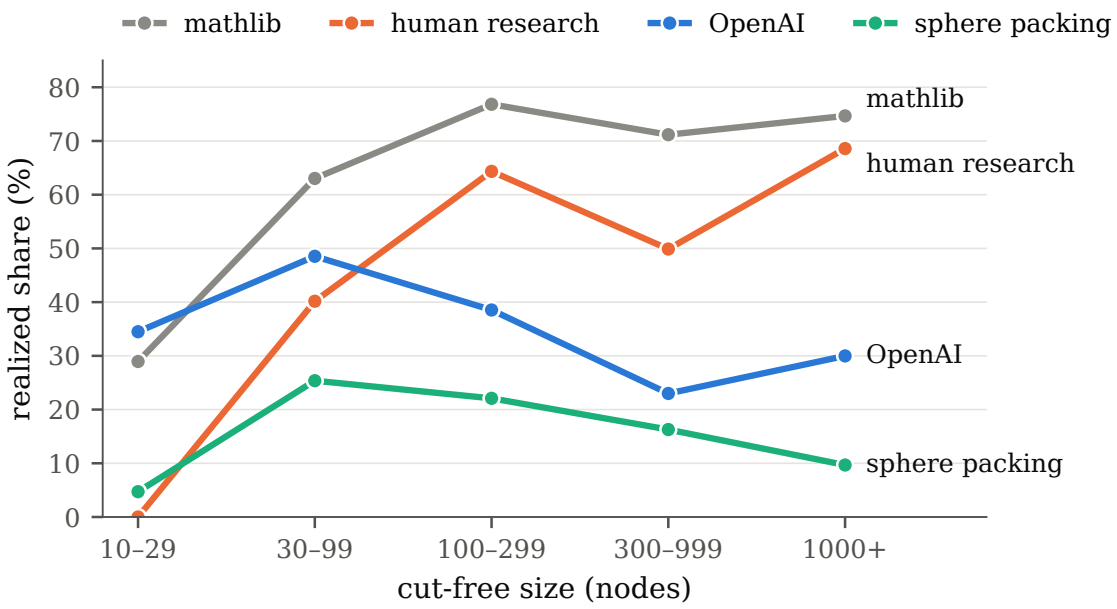


Figure 5: Realized share within bins of cut-free proof size. The bins hold 89,144, 42,600, 8,393, 2,264, and 906 mathlib proofs; 10,301, 9,219, 3,148, 2,121, and 1,049 OpenAI proofs; 1,402 to 115 human proofs; and 1,263 to 285 sphere packing proofs. Putnam and IMO are left out, since most of their bins hold fewer than 30 proofs.

than mathlib's with an AUC of 0.25 ($p = 6 \cdot 10^{-15}$), and of 0.36 on proofs that call no automation. OpenAI therefore loses in both directions, even on proofs that call no automation. It writes cuts that save nothing (the redundancy ρ), and it leaves out cuts that would save something.

Mathlib also leaves compression unrealized. Across all its proofs, the author's cuts reduce the cut-free size by 36%, the explicit layout by 48%, and full sharing by 55%. Its largest misses are computational. The proof that the Cartan matrix of E_8 has determinant 1 has a cut-free size of 561,653 nodes, and its author's layout saves nothing. An explicit layout would bring it to 21,052 nodes, and full sharing to 8,156. The same holds for E_6 and E_7 and for several formulas on Weierstrass curves. In each case a tactic emits the same subproof many times.

Two subproofs are counted as shared only if they are identical. A mathematician's lemma is usually a *general* statement, with a parameter, used at several different instances. Such lemmas are invisible to this measure, so realized compression captures repetition only. Abstraction of this kind is the target of the literature on cut *introduction* (e.g. the work of Hetzl, Leitsch, Reis, and Weller *(Hetzl et al. 2012; Hetzl et al. 2014a)*). Measuring it at this scale remains open.

2.14 Is it only tooling?

We asked above whether the divergence is only layout, and the same question can be asked of the tools. As noted among the confounds, a tactic such as `ring` or `omega` writes a certificate into the proof term, and the certificate has its own gestures. Certificates make up 32% of OpenAI's gestures and 9% of mathlib's. The rewriting that `rw` and `simp` leave behind (chains of `Eq.mpr`, `congrArg`, and `Eq.trans`) is the same kind of artifact. To check, I collapsed each certificate into a single step, which records that a tactic was called, and kept everything else, including the facts the author passes to the tactic. I then reran the entire gesture analysis (vocabulary, eigenproofs, null) three times: with automation collapsed, with automation and rewriting collapsed, and with both removed, so that only the author's own steps remain.

OpenAI's divergence falls from 280 to 182 bits, against a null of 49 (3.7 times). With rewriting collapsed as well, it is 191 bits against 56 (3.4 times), and on the author's own steps, 220 bits against 56 (3.9 times). Each of these is far above the largest value the null produced (60 to 68 bits). So about a third of the divergence comes from the tools, and the rest does not. Human research code behaves differently: its divergence falls from 1.8 to 1.1 times its null, within the range of the null itself, so most of what separated it from mathlib was tooling. For OpenAI, with automation collapsed, the first eigenproof is how often a tactic is called, together with lemmas about inequalities. On the author's own steps, it is chains of inequality lemmas (from $\leq$ to $\leq$, and from $<$ to $\leq$). This may be subject matter rather than style, since the gesture study does not compare within areas.

For the other measures, the simplest control is to keep only the proofs that call no automation at all. OpenAI calls automation in 21% of its proofs, against 4% of mathlib's, but it rewrites about as much. The AUCs of OpenAI against mathlib, on all proofs and on proofs that call no automation, are (proofs of at least 10 steps, in projects with at least 20 such proofs, so the first column differs slightly from the earlier tables):

	all proofs	no automation
steps per theorem	0.87	0.73
unused hypotheses	0.75	0.77
intermediate claims per 100 steps	0.94	0.95
redundancy ρ	0.95	0.94
compressibility χ	0.63	0.55
κ	0.12	0.07
realized share	0.25	0.36

The redundant claims, the unused hypotheses, and the low κ do not depend on the tools. OpenAI's proofs

are longer mostly because they call more automation, and the slight excess in χ comes from automation alone ($p = 0.14$ without it), since certificates repeat themselves. The realized share moves the most, because certificates hold most of the compression available in the proofs that call automation, and no author can name a subproof inside one. On proofs that call no automation, mathlib realizes 86% of the available compression, human research code 82%, OpenAI 55%, and sphere packing 33%. OpenAI still realizes less than mathlib (AUC 0.36, $p = 1 \cdot 10^{-5}$) and less than human research code (AUC 0.28, $p = 0.01$), but the gap is smaller than over all proofs, and human research code no longer differs from mathlib (AUC 0.49).

2.15 Checking the instruments

Some of what our instruments should report is fixed by theorems. A measurement that violates a theorem is a broken measurement, so I treat the following results as calibration.

First, by the reading of κ above, a step can be copied at most as often as the product of the uses of the cuts around it. This gives two bounds:

$$\kappa \le \sum_{\text{cuts}} \log_2(\text{uses}), \qquad \chi_{\text{DAG}} \le \log_2(\text{most copies of any subproof}).$$

Both hold for every proof in every corpus. The first bound is loose (in mathlib, κ is typically 0.18 of it), since it ignores how large the copied proofs are. A refined estimate that accounts for size, $\sum \log_2(1+(u-1)\cdot|p|/N)$ over cuts (with u the cut's uses, $|p|$ the size of its proof, and N the written size), tracks κ with a Spearman rank correlation *(Spearman 1904)* of 0.95 in mathlib.

Second, the enormous blowups found by Statman and Orevkov have a known source. A *redex* is a function applied to an argument, which is how a cut appears in a proof term. When a cut is eliminated, a proof can be substituted into a position where it is applied to another proof. This produces a new redex, called a *created redex*. Define the *rank* of a cut as the depth of implications nested to the left in its formula (with $\neg A$ read as $A \to \bot$). A created redex can only arise from a cut of rank at least one. Cuts of rank zero cost at most a single exponential when eliminated. I call this the *rank ladder*.

When I first checked the rank ladder, however, it failed for more than half of the mathlib proofs with created redexes. A theorem cannot fail, so the instrument had to be wrong. By printing a counterexample, I found two errors. The pass read ranks from the written formula, so that $s \subseteq t$ or `Monotone f`, which are implications once their definitions are unfolded, counted as rank zero. It also missed cuts hidden behind substitutions of data. Once both were fixed, the ladder held for 99.6% of 1,100 mathlib proofs with created redexes. I printed the five remaining exceptions (four in mathlib, one in the IMO corpus), and each had a

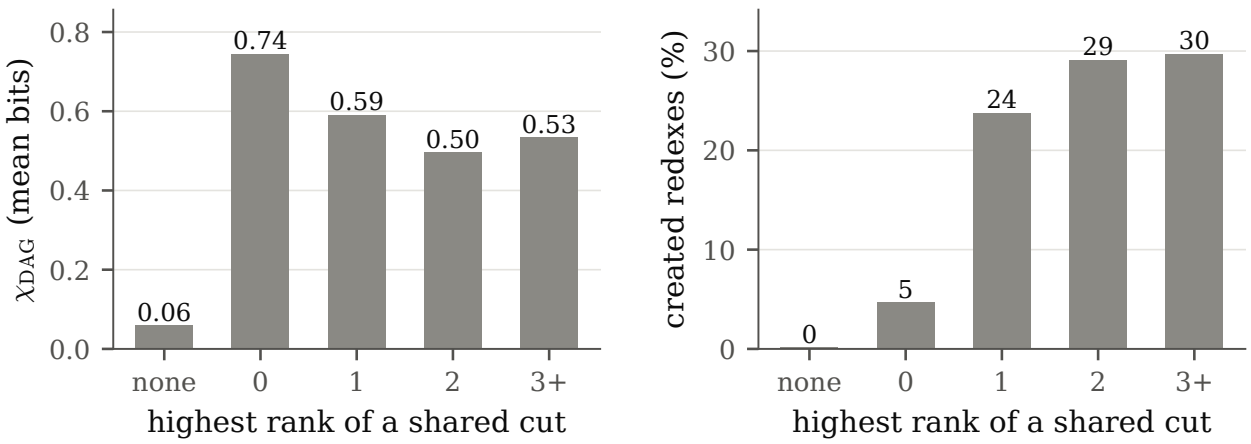


Figure 6: Mathlib proofs grouped by the highest logical rank among their shared cuts (definitions unfolded): 134,172 with no shared cut, then 6,752, 1,992, 385, and 138. Left: available compression. Right: share of proofs in which eliminating cuts creates new proof redexes.

cut of rank one that the pass read as rank zero. Two cuts are inclusions of submodules, $p \leq q$, and one is an inclusion of finite sets; their orders are built with `let`, which the unfolding did not pass through. One cut's hypothesis is a proposition only through a defined family (the arrows of a sieve), which the pass unfolded before reading its type. The last mentions a set defined by a local `let`, which the pass did not substitute. With these fixed, the ladder holds for every proof with created redexes in every corpus (1,100 in mathlib, and 35, 492, 50, 14, and 5 in the human, OpenAI, sphere packing, Putnam, and IMO corpora).

Third, real libraries are far from the worst case. Sharing could halve a proof ($\chi_{\mathrm{DAG}} \geq 1$) in only 2.1% of mathlib proofs. Created redexes appear in 0.8% of mathlib proofs and 1.9% of OpenAI's. With definitions unfolded, 21% of mathlib's cuts have rank at least one (against 8% when ranks are read syntactically). In mathlib, proofs whose shared cuts reach rank one or more have *less* compression available than those with only rank-zero shared cuts (χ_{DAG} of 0.50 to 0.59, against 0.74; Figure 6). What rises with rank is created redexes, from 5% to 30%. A shared cut of rank one is a local lemma of the form $A \to B$ that is applied at several places. Positive rank is the precondition for Boolos-type speed-ups. In mathlib, the proofs that contain positive-rank shared cuts are less compressible than proofs whose shared cuts all have rank zero, which suggests that such cuts serve abstraction rather than compression.

Finally, the pipeline was checked against itself. The driver's first run and the hashing pass compute cut-free sizes independently, the first by normalizing each proof and the second by hashing its term. They agree for 98.5% of mathlib proofs, and the exceptions are the created redexes, which the first run reduces and the hashing pass leaves in place. The OpenAI projects compiled both ways (see "Preparing the corpora") also give the same proof terms: of 2,879 proofs from seven of them, 99.8% have identical sizes, and the 5 that

differ do so outside their cuts.

3 From mathematical data to mathematical evidence?

Tycho Brahe's observations of the heavens were valuable because their precision made discrepancies impossible to ignore. Their provenance mattered too, in the long run especially: Kepler inherited measurements that could be revisited, recalculated, and brought to bear on competing models. It would be rather grandiose to cast our collection of Lean files as a new observatory on Hven, and I do not mean to suggest that we are on the verge of a Newtonian synthesis. Still, a compiled Lean file gives us something unusually definite: a proof term checked by the kernel against a proposition in a specified formal environment. This establishes that the term proves the proposition Lean has elaborated, relative to the declarations and axioms in play. This may be less than it seems: the formal proposition may fail to capture the intended mathematical claim; the accompanying informal argument may say something different; the proof may depend on a particular library version or environment that others cannot readily reproduce. If these artifacts are to become useful mathematical data, we must keep track of where they came from, what they depend on, and which attempts to compile them failed.

OpenAI's public mathematics repository (whose openness was crucial to the success of this project) contains hundreds of manuscripts grouped into families, while its README *(OpenAI 2026b)* acknowledges that the results occupy different stages of verification and that roughly 42% of the top-line results have Lean formalizations. Moreover, on 6 October 2026, Bastounis, Circelli, and Hansen *(Bastounis et al. 2026)* published a preprint arguing that parts of OpenAI's announced Navier–Stokes proof do not correspond faithfully to their Lean formalizations. One of their examples concerns an inverse estimate for which the informal paper requires four additional derivatives while the cited formal result uses five; the authors also identify discrepancies in a pressure-flux estimate and its proof. The authors distinguish their claims about formalization mismatches from a judgment on the correctness of the informal proof. Kallel and El Louadi make a related argument from another release of AI-written results, in which the statements a human must audit are a small fraction of the kernel-checked proof *(Kallel and El Louadi 2026)*. These criticisms illustrate a rather immediate difficulty for empirical work: a formal verification result can be sound relative to its environment while failing to certify the claim for which it is presented as evidence.

The problem becomes even more acute when such artifacts are collected at scale. A corpus that treats compiled proofs, uncompiled drafts, incomplete formalizations, and proofs checked under incompatible environments as equivalent observations will obscure precisely the distinctions it is meant to investigate. Opaque

or redundant derivations also impose costs on subsequent researchers, who must recover their dependencies, determine what has actually been established, and distinguish mathematical structure from accidental features of the generation or formalization process. The opposite policy, discarding every proof that fails to compile against a chosen library version, introduces its own bias. In the corpora examined here, moving to a common Lean and mathlib environment eliminates 20% of the sphere-packing theorems, 11% of the Putnam solutions, and 67% of the IMO solutions; 222 of the 4,898 theorems in the human-written Carleson project also fail to survive the migration. Longer proofs are more likely to break, so the surviving corpus systematically favors shorter ones. A dataset assembled without recording these losses could therefore mistake compatibility with a particular software environment for a structural property of mathematical reasoning.

This is where empirical analysis becomes part of the solution. In the present study, the rank-ladder test initially failed on more than half of the mathlib proofs to which it applied. Since the bound follows mathematically, these failures exposed defects in the measurement procedure. Inspecting the counterexamples revealed errors in the treatment of defined propositions and substitutions; correcting the instrument restored the expected bound in every tested case. The same principle extends to the corpus itself: unusual proof structures, unexplained changes across versions, excessive intermediate claims, and abnormal patterns of sharing can identify artifacts that merit closer inspection. Of course, such analyses cannot replace checking the correspondence between a formal statement and its intended mathematical meaning. They can, however, make that audit more systematic and help identify where the underlying data or the instruments used to study it have gone wrong. In my ideal world, the aim would be to build a mathematical archive in which failures and corrections remain visible, so that accumulated proofs become a resource for discovering regularities rather than a mass of outputs whose evidential status is impossible to recover.

Most abstractly, underlying this enterprise is the Siren's call to Cantor's paradise *(Hilbert 1926)*. Once a proof has been formalized, it becomes possible to collect it alongside thousands of others, strip away much of its mathematical content, and compare structures that would otherwise be difficult to place on common footing. This is what the notion of a gesture attempts to do. We erase names, retain the head connective of each type, and record how often each hypothesis is used. The result is a description of a proof that can be compared with proofs of entirely different theorems. Naturally, we lose something in the process; Frege would have his objection ready. Even so, the abstraction separates the corpora: OpenAI's gesture distribution lies 280 bits from mathlib's, against 49 bits for a typical size-matched sample of mathlib compared with itself. The divergence persists when we restrict attention to proofs in their optimal layout, and when each call to automation is collapsed into a single step. We have constructed a common space in which a difference

between collections of proofs becomes measurable, even though the theorems they establish may have little in common.

But, the Kroneckerian impulse enters at the point where we insist on seeing the construction itself. A theorem statement in a manuscript is one thing; a proof term that Lean's kernel has checked is another. This insistence imposes costs: to compare the corpora, I compiled them against a common version of Lean and mathlib, and some proofs did not survive the journey. Moreover, longer proofs were more likely to fail, so the surviving data were biased towards shorter proofs. Here the demand for an effectively exhibited construction collides with the ambition to study mathematics in its full variety. We cannot simply declare every surviving proof representative of its original corpus, nor can we conclude that every discarded proof was mathematically defective. Some failures reflect changes in libraries and toolchains. The exactness of our criterion does not make the selection it imposes innocuous–and we can always hunt for more fine-grained ways to classify our proofs.

There is a similar tension that appears within the proof terms themselves. We can ask how much a proof contracts when its intermediate claims are eliminated, and how much repeated structure its author has managed to exploit. These questions admit exact answers for the formal objects we study, even when normalization would produce a term far too large to write down explicitly. Yet the answers resist any simple celebration of machine-generated mathematics–OpenAI's proofs contain more intermediate claims than either mathlib or the human research projects, but those claims often do little work: 40% of OpenAI's proofs are larger with their cuts than without them, against 13% in mathlib and 26% in the human research corpus. OpenAI also realizes only 31% of the compression available through repeated subproofs, against 73% in mathlib (and 55% against 86% on proofs that call no automation). Of course, these are merely properties of the derivations we measured, not verdicts on the mathematical worth of their conclusions. But then the whole point of the exercise is precisely that such distinctions can be made, tested, and revised. The present archive gives us something to investigate, and we can investigate it with eyes towards its limitations.

To me, the most interesting prospect of this line of inquiry is that this study will never quite finish collecting its object. A proof changes its appearance as we move between source text, elaborated term, cut-free form, and compressed representation; a library changes beneath it, and an apparently innocuous change in our instruments can send us back to reconsider what we thought we had measured. This makes the empirical study of proofs a rather peculiar science, whose data are themselves mathematical constructions and whose instruments must answer to mathematical theorems. There is something here that vindicates Imre Lakatos's insights in *Proofs and Refutations (Lakatos 1976)*: the proof is a moving object, altered by criticism, counterexample, and the introduction of distinctions that were invisible in the original formulation. Perhaps,

as these corpora grow, we will discover regularities in mathematical reasoning that are difficult to see from within any single proof, or find that some of our present categories collapse under closer inspection. Either way, the study will have to proceed through conjectures about proofs, counterexamples to our conjectures, and revisions to the concepts with which we describe them. The archive will change as mathematics changes, and our account of mathematical practice will have to change with it.

References

Achim, Tudor et al. (2025). *Aristotle: IMO-level automated theorem proving*. arXiv: 2510.01346.

Ahuja, Riyaz et al. (2024). *ImProver: Agent-based automated proof optimization*. arXiv: 2410.04753.

Anderson, Aaron et al. (2023). *Formalization of the polynomial Freiman–Ruzsa conjecture of Marton*. Commit ce7ac95. URL: https://github.com/teorth/pfr.

Anthropic (2026). *Claude Code*. URL: https://claude.com/product/claude-code.

Aryal, Safal R. (2026). *math-ai-projects: Code and results for "Proofs, gestures, and formalization"*. URL: https://github.com/safsom/math-ai-projects.

Aspinall, David and Cezary Kaliszyk (2016). "Towards formal proof metrics". In: *Fundamental Approaches to Software Engineering (FASE 2016)*. Berlin: Springer, pp. 325–341. DOI: 10.1007/978-3-662-49665-7_19.

Axiom Math (2026a). *AxiomProver at Putnam 2025*. Commit 2653cde. URL: https://github.com/AxiomMath/Putnam2025.

Bastounis, Alexander, Fabian Circelli, and Anders C. Hansen (2026). *Navier-Stokes lost in translation: Why Lean verification of AI autoformalisation does not guarantee correct natural language proofs*. Submitted 6 October 2026. arXiv: 2610.08144.

Becker, Lars et al. (2024). *A blueprint for the formalization of Carleson's theorem on convergence of Fourier series*. arXiv: 2405.06423.

Blanchette, Jasmin Christian et al. (2015). "Mining the Archive of Formal Proofs". In: *Intelligent Computer Mathematics (CICM 2015)*. Cham: Springer, pp. 3–17. DOI: 10.1007/978-3-319-20615-8_1.

Blumenthal, Otto (1935). "Lebensgeschichte". In: *David Hilbert: Gesammelte Abhandlungen*. Vol. 3. Berlin: Julius Springer, pp. 388–429.

Boolos, George (1984). "Don't eliminate cut". In: *Journal of Philosophical Logic* 13.4, pp. 373–378. DOI: 10.1007/BF00247711.

Bourigault, Pauline (2026). *LeanPolish: Verified supervision for Lean proof compression*. arXiv: 2609.38384.

Brahe, Tycho (1598). *Astronomiae instauratae mechanica*. Wandesburgi [Wandsbek].

Brouwer, L. E. J. (1907). "Over de grondslagen der wiskunde". PhD thesis. Amsterdam: Universiteit van Amsterdam.

— (1918). "Begründung der Mengenlehre unabhängig vom logischen Satz vom ausgeschlossenen Dritten. Erster Teil: Allgemeine Mengenlehre". In: *Verhandelingen der Koninklijke Akademie van Wetenschappen te Amsterdam, Eerste Sectie* 12.5.

— (1952). "Historical background, principles and methods of intuitionism". In: *South African Journal of Science* 49, pp. 139–146.

— (1975). *Collected Works*. Ed. by Arend Heyting. Vol. 1: Philosophy and Foundations of Mathematics. Amsterdam: North-Holland.

Buzzard, Kevin and Richard Taylor (2025). *FLT: An ongoing Lean formalization of Fermat's Last Theorem*. Commit f6c6a2f. URL: https://github.com/ImperialCollegeLondon/FLT.

Caldwell, Shane (2026). *ProofJudge: Tool-grounded LLM evaluation of formal proof quality in Mathlib*. arXiv: 2608.20432.

Cantor, Georg (1883a). *Grundlagen einer allgemeinen Mannichfaltigkeitslehre: Ein mathematisch-philosophischer Versuch in der Lehre des Unendlichen*. Leipzig: B. G. Teubner.

Chen, Luoxin et al. (2025). *Seed-Prover: Deep and broad reasoning for automated theorem proving*. arXiv: 2507.23726.

Chen, Stanley F. and Joshua Goodman (1999). "An empirical study of smoothing techniques for language modeling". In: *Computer Speech & Language* 13.4, pp. 359–393. DOI: 10.1006/csla.1999.0128.

Coquand, Thierry and Gérard Huet (1988). "The calculus of constructions". In: *Information and Computation* 76.2–3, pp. 95–120. DOI: 10.1016/0890-5401(88)90005-3.

Curry, Haskell B. and Robert Feys (1958). *Combinatory Logic*. Vol. 1. Amsterdam: North-Holland.

De Moura, Leonardo and Sebastian Ullrich (2021). "The Lean 4 theorem prover and programming language". In: *Automated Deduction – CADE 28*. Cham: Springer, pp. 625–635. DOI: 10.1007/978-3-030-79876-5_37.

Ebner, Gabriel et al. (2018). "On the generation of quantified lemmas". In: *Journal of Automated Reasoning* 63.1, pp. 95–126. DOI: 10.1007/s10817-018-9462-8.

Euclid (1908). *The Thirteen Books of Euclid's Elements*. Trans. by Thomas L. Heath. 3 vols. Cambridge: Cambridge University Press.

Filliâtre, Jean-Christophe and Sylvain Conchon (2006). "Type-safe modular hash-consing". In: *Proceedings of the 2006 Workshop on ML*. New York: ACM, pp. 12–19. DOI: 10.1145/1159876.1159880.

Frege, Gottlob (1884). *Die Grundlagen der Arithmetik: Eine logisch mathematische Untersuchung über den Begriff der Zahl*. Breslau: Wilhelm Koebner.

Frege, Gottlob (1950). *The Foundations of Arithmetic: A Logico-Mathematical Enquiry into the Concept of Number*. Trans. by J. L. Austin. Oxford: Basil Blackwell.

— (1980). *Philosophical and Mathematical Correspondence*. Ed. by Gottfried Gabriel et al. Trans. by Hans Kaal. Abridged from the German edition by Brian McGuinness. Oxford: Basil Blackwell.

Fu, Yiming et al. (2026). *Proof-Refactor: Refactoring generated formal proofs into modular artifacts*. arXiv: 2606.03743.

Galilei, Galileo (1989). *Sidereus Nuncius, or The Sidereal Messenger*. Trans. by Albert Van Helden. First published Venice, 1610. Chicago: University of Chicago Press.

Galilei, Galileo and Christoph Scheiner (2010). *On Sunspots*. Trans. by Eileen Reeves and Albert Van Helden. Chicago: University of Chicago Press.

Gentzen, Gerhard (1935a). "Untersuchungen über das logische Schließen. I". In: *Mathematische Zeitschrift* 39.1, pp. 176–210. DOI: 10.1007/BF01201353.

— (1935b). "Untersuchungen über das logische Schließen. II". In: *Mathematische Zeitschrift* 39.1, pp. 405–431. DOI: 10.1007/BF01201363.

Gu, Alex et al. (2025). *ProofOptimizer: Training language models to simplify proofs without human demonstrations*. arXiv: 2510.15700.

Hadamard, Jacques (1905). "Cinq lettres sur la théorie des ensembles". In: *Bulletin de la Société Mathématique de France* 33. Letters exchanged between Baire, Borel, Hadamard, and Lebesgue, pp. 261–273. DOI: 10.24033/bsmf.761.

Han, Jesse Michael et al. (2021). *Proof artifact co-training for theorem proving with language models*. arXiv: 2102.06203.

Hanley, James A. and Barbara J. McNeil (1982). "The meaning and use of the area under a receiver operating characteristic (ROC) curve". In: *Radiology* 143.1, pp. 29–36. DOI: 10.1148/radiology.143.1.7063747.

Hariharan, Sidharth et al. (2026). *Progress in formalizing sphere packing in dimension 8*. arXiv: 2604.23468.

Harmonic (2025). *Aristotle IMO 2025 solutions in Lean*. Commit 72b6240. URL: https://github.com/harmonic-ai/IMO2025.

Hetzl, Stefan, Alexander Leitsch, and Daniel Weller (2012). "Towards algorithmic cut-introduction". In: *Logic for Programming, Artificial Intelligence, and Reasoning (LPAR-18)*. Berlin: Springer, pp. 228–242. DOI: 10.1007/978-3-642-28717-6_19.

Hetzl, Stefan et al. (2014a). "Algorithmic introduction of quantified cuts". In: *Theoretical Computer Science* 549, pp. 1–16. DOI: 10.1016/j.tcs.2014.05.018.

Heyting, Arend (1930a). "Die formalen Regeln der intuitionistischen Logik". In: *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse*, pp. 42–56.

Heyting, Arend (1930b). "Die formalen Regeln der intuitionistischen Mathematik II". In: *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse*, pp. 57–71.

— (1930c). "Die formalen Regeln der intuitionistischen Mathematik III". In: *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse*, pp. 158–169.

Hilbert, David (1899). "Grundlagen der Geometrie". In: *Festschrift zur Feier der Enthüllung des Gauss-Weber-Denkmals in Göttingen*. Leipzig: B. G. Teubner, pp. 1–92.

— (1926). "Über das Unendliche". In: *Mathematische Annalen* 95.1, pp. 161–190. DOI: 10.1007/BF01206605.

— (1928). "Die Grundlagen der Mathematik". In: *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg* 6.1. Lecture given in Hamburg, July 1927, pp. 65–85. DOI: 10.1007/BF02940602.

— (1967). "The foundations of mathematics". In: *From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931*. Ed. by Jean van Heijenoort. Trans. by Stefan Bauer-Mengelberg and Dagfinn Føllesdal. Cambridge, MA: Harvard University Press, pp. 464–479.

Hindle, Abram et al. (2012). "On the naturalness of software". In: *Proceedings of the 34th International Conference on Software Engineering (ICSE)*. IEEE, pp. 837–847. DOI: 10.1109/ICSE.2012.6227135.

Howard, William A. (1980). "The formulae-as-types notion of construction". In: *To H. B. Curry: Essays on Combinatory Logic, Lambda Calculus and Formalism*. Ed. by J. P. Seldin and J. R. Hindley. London: Academic Press, pp. 479–490.

Hubert, Thomas et al. (2025). "Olympiad-level formal mathematical reasoning with reinforcement learning". In: *Nature* 651.8106, pp. 607–613. DOI: 10.1038/s41586-025-09833-y.

Kaliszyk, Cezary and Josef Urban (2013). "Lemma mining over HOL Light". In: *Logic for Programming, Artificial Intelligence, and Reasoning (LPAR-19)*. Berlin: Springer, pp. 503–517. DOI: 10.1007/978-3-642-45221-5_34.

Kallel, Maher and Mohamed El Louadi (2026). *Verification abundance, adjudication scarcity: What happens to mathematical knowledge when proof checking becomes free*. arXiv: 2608.28997.

Kepler, Johannes (1609). *Astronomia nova*. Heidelberg.

— (1992). *New Astronomy*. Trans. by William H. Donahue. Cambridge: Cambridge University Press.

Kneser, Reinhard and Hermann Ney (1995). "Improved backing-off for M-gram language modeling". In: *1995 International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. Vol. 1. IEEE, pp. 181–184. DOI: 10.1109/ICASSP.1995.479394.

Kullback, S. and R. A. Leibler (1951). "On information and sufficiency". In: *The Annals of Mathematical Statistics* 22.1, pp. 79–86. DOI: 10.1214/aoms/1177729694.

Lakatos, Imre (1976). *Proofs and Refutations: The Logic of Mathematical Discovery*. Ed. by John Worrall and Elie Zahar. Cambridge: Cambridge University Press.

Lin, Yong et al. (2025). *Goedel-Prover: A frontier model for open-source automated theorem proving*. arXiv: 2502.07640.

Mann, H. B. and D. R. Whitney (1947). "On a test of whether one of two random variables is stochastically larger than the other". In: *The Annals of Mathematical Statistics* 18.1, pp. 50–60. DOI: 10.1214/aoms/1177730491.

Massot, Patrick, Oliver Nash, and Floris van Doorn (2026). *The sphere eversion project*. Commit df3e74a. URL: https://github.com/leanprover-community/sphere-eversion.

Math, Inc., Sidharth Hariharan, Maryna Viazovska, et al. (2026). *Sphere packing in dimensions 8 and 24 in Lean (completed by the Gauss autoformalization agent)*. Commit 1e98fb4. URL: https://github.com/math-inc/Sphere-Packing-Lean.

Newton, Isaac (1687). *Philosophiæ Naturalis Principia Mathematica*. London: Jussu Societatis Regiæ ac Typis Josephi Streater.

— (1999). *The Principia: Mathematical Principles of Natural Philosophy*. Trans. by I. Bernard Cohen and Anne Whitman. Berkeley: University of California Press.

OpenAI (Sept. 8, 2026a). *On the Navier–Stokes Millennium Prize problem*. URL: https://openai.com/index/navier-stokes-solution/.

— (2026b). *openai/math: Mathematical manuscripts and Lean formalizations produced by an OpenAI model*. Commit fd4aeeb2e. URL: https://github.com/openai/math.

Orevkov, V. P. (1982). "Lower bounds for increasing complexity of derivations after cut elimination". In: *Journal of Soviet Mathematics* 20.4, pp. 2337–2350. DOI: 10.1007/BF01629444.

Pugh, William (1991). "The Omega test: A fast and practical integer programming algorithm for dependence analysis". In: *Proceedings of the 1991 ACM/IEEE Conference on Supercomputing*. New York: ACM, pp. 4–13. DOI: 10.1145/125826.125848.

Ren, Z. Z. et al. (2025). *DeepSeek-Prover-V2: Advancing formal mathematical reasoning via reinforcement learning for subgoal decomposition*. arXiv: 2504.21801.

Spearman, C. (1904). "The proof and measurement of association between two things". In: *The American Journal of Psychology* 15.1, pp. 72–101. DOI: 10.2307/1412159.

Statman, Richard (1979a). "Lower bounds on Herbrand's theorem". In: *Proceedings of the American Mathematical Society* 75.1, pp. 104–107. DOI: 10.1090/S0002-9939-1979-0529224-9.

— (1979b). "The typed λ-calculus is not elementary recursive". In: *Theoretical Computer Science* 9.1, pp. 73–81. DOI: 10.1016/0304-3975(79)90007-0.

Stephenson, Bruce (1987). *Kepler's Physical Astronomy*. Studies in the History of Mathematics and Physical Sciences 13. New York: Springer.

The mathlib Community (2020). "The Lean mathematical library". In: *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*. New York: ACM, pp. 367–381. DOI: 10.1145/3372885.3373824.

— (2026). *mathlib4*. Commit d13f23b. URL: https://github.com/leanprover-community/mathlib4.

Thoren, Victor E. (1990). *The Lord of Uraniborg: A Biography of Tycho Brahe*. Cambridge: Cambridge University Press.

Troelstra, A. S. and H. Schwichtenberg (2000). *Basic Proof Theory*. 2nd ed. Cambridge Tracts in Theoretical Computer Science 43. Cambridge: Cambridge University Press.

Van Doorn, Floris et al. (2026). *Formalization of a generalized Carleson's theorem*. Commit 8972acd. URL: https://github.com/fpvandoorn/carleson.

Vyskočil, Jiří, David Stanovský, and Josef Urban (2010). "Automated proof compression by invention of new definitions". In: *Logic for Programming, Artificial Intelligence, and Reasoning (LPAR-16)*. Berlin: Springer, pp. 447–462. DOI: 10.1007/978-3-642-17511-4_25.

Wang, Haiming et al. (2025). *Kimina-Prover preview: Towards large formal reasoning models with reinforcement learning*. arXiv: 2504.11354.

Weber, Heinrich (1893). "Leopold Kronecker". In: *Jahresbericht der Deutschen Mathematiker-Vereinigung* 2, pp. 5–31.

Whiteside, D. T., ed. (1974). *The Mathematical Papers of Isaac Newton*. Vol. 6: 1684–1691. Contains Newton's *De motu corporum in gyrum* (1684). Cambridge: Cambridge University Press.

Whiteside, Iain et al. (2011). "Towards formal proof script refactoring". In: *Intelligent Computer Mathematics (CICM 2011)*. Berlin: Springer, pp. 260–275. DOI: 10.1007/978-3-642-22673-1_18.

Wiedijk, Freek (2000). *The de Bruijn factor*. Poster at TPHOLs 2000. URL: https://www.cs.ru.nl/~freek/factor/factor.pdf.

Further related work

Axiom Math (Jan. 8, 2026b). *From seeing why to checking everything*. URL: https://axiommath.ai/research/from-seeing-why-to-checking-everything.

Baaz, Matthias and Alexander Leitsch (2010). "Complexity of cut-elimination". In: *Methods of Cut-Elimination*. Dordrecht: Springer, pp. 39–61. DOI: 10.1007/978-94-007-0320-9_4.

Beckmann, Arnold (2001). "Exact bounds for lengths of reductions in typed λ-calculus". In: *Journal of Symbolic Logic* 66.3, pp. 1277–1285. DOI: 10.2307/2695106.

Cantor, Georg (1883b). "Ueber unendliche, lineare Punktmannichfaltigkeiten". In: *Mathematische Annalen* 21.4. Part 5; also issued separately as *Grundlagen einer allgemeinen Mannichfaltigkeitslehre*, pp. 545–591. DOI: 10.1007/BF01446819.

Charikar, Moses et al. (2005). "The smallest grammar problem". In: *IEEE Transactions on Information Theory* 51.7, pp. 2554–2576. DOI: 10.1109/TIT.2005.850116.

Cohn, Henry et al. (2017). "The sphere packing problem in dimension 24". In: *Annals of Mathematics* 185.3, pp. 1017–1033. DOI: 10.4007/annals.2017.185.3.8.

Downey, Peter J., Ravi Sethi, and Robert Endre Tarjan (1980). "Variations on the common subexpression problem". In: *Journal of the ACM* 27.4, pp. 758–771. DOI: 10.1145/322217.322228.

Eberhard, Sebastian, Gabriel Ebner, and Stefan Hetzl (2017). "Algorithmic compression of finite tree languages by rigid acyclic grammars". In: *ACM Transactions on Computational Logic* 18.4, pp. 1–20. DOI: 10.1145/3127401.

Eberhard, Sebastian and Stefan Hetzl (2015a). "Compressibility of finite languages by grammars". In: *Descriptional Complexity of Formal Systems (DCFS 2015)*. Cham: Springer, pp. 93–104. DOI: 10.1007/978-3-319-19225-3_8.

— (2015b). "Inductive theorem proving based on tree grammars". In: *Annals of Pure and Applied Logic* 166.6, pp. 665–700. DOI: 10.1016/j.apal.2015.01.002.

— (2018). "On the compressibility of finite languages and formal proofs". In: *Information and Computation* 259, pp. 191–213. DOI: 10.1016/j.ic.2017.09.001.

Ebner, Gabriel et al. (2016). "System description: GAPT 2.0". In: *Automated Reasoning (IJCAR 2016)*. Cham: Springer, pp. 293–301. DOI: 10.1007/978-3-319-40229-1_20.

Heras, Jónathan and Ekaterina Komendantskaya (2014). "Recycling proof patterns in Coq: Case studies". In: *Mathematics in Computer Science* 8.1, pp. 99–116. DOI: 10.1007/s11786-014-0173-1.

Hetzl, Stefan (2012). "Applying tree languages in proof theory". In: *Language and Automata Theory and Applications (LATA 2012)*. Berlin: Springer, pp. 301–312. DOI: 10.1007/978-3-642-28332-1_26.

Hetzl, Stefan et al. (2014b). "Introducing quantified cuts in logic with equality". In: *Automated Reasoning (IJCAR 2014)*. Cham: Springer, pp. 240–254. DOI: 10.1007/978-3-319-08587-6_17.

Ilin, Vasily (2026). *Lean Pool: An AI-maintained archive of formalized mathematics*. arXiv: 2609.25199.

Kaliszyk, Cezary and Josef Urban (2015). "Learning-assisted theorem proving with millions of lemmas". In: *Journal of Symbolic Computation* 69, pp. 109–128. DOI: 10.1016/j.jsc.2014.09.032.

Kaliszyk, Cezary, Josef Urban, and Jiří Vyskočil (2015). "Lemmatization for stronger reasoning in large theories". In: *Frontiers of Combining Systems (FroCoS 2015)*. Cham: Springer, pp. 341–356. DOI: 10.1007/978-3-319-24246-0_21.

Lample, Guillaume et al. (2022). *HyperTree proof search for neural theorem proving*. arXiv: 2205.11491.

Ledoit, Olivier and Michael Wolf (2004). "A well-conditioned estimator for large-dimensional covariance matrices". In: *Journal of Multivariate Analysis* 88.2, pp. 365–411. DOI: 10.1016/S0047-259X(03)00096-4.

Leitsch, Alexander and Michael Lettmann (2018). "The problem of Π_2-cut-introduction". In: *Theoretical Computer Science* 706, pp. 83–116. DOI: 10.1016/j.tcs.2017.10.003.

Lohrey, Markus (2012). "Algorithmics on SLP-compressed strings: A survey". In: *Groups – Complexity – Cryptology* 4.2. DOI: 10.1515/gcc-2012-0016.

Polu, Stanislas and Ilya Sutskever (2020). *Generative language modeling for automated theorem proving*. arXiv: 2009.03393.

Pudlák, Pavel (1998). "The lengths of proofs". In: *Handbook of Proof Theory*. Ed. by Samuel R. Buss. Vol. 137. Studies in Logic and the Foundations of Mathematics. Amsterdam: Elsevier, pp. 547–637. DOI: 10.1016/S0049-237X(98)80023-2.

Schwichtenberg, Helmut (1982). "Complexity of normalization in the pure typed lambda-calculus". In: *The L. E. J. Brouwer Centenary Symposium*. Vol. 110. Studies in Logic and the Foundations of Mathematics. Amsterdam: North-Holland, pp. 453–457. DOI: 10.1016/S0049-237X(09)70143-0.

Tsoukalas, George et al. (2024). *PutnamBench: Evaluating neural theorem-provers on the Putnam mathematical competition*. arXiv: 2407.11214.

Viazovska, Maryna (2017). "The sphere packing problem in dimension 8". In: *Annals of Mathematics* 185.3, pp. 991–1015. DOI: 10.4007/annals.2017.185.3.7.

Xin, Huajian et al. (2024a). *DeepSeek-Prover-V1.5: Harnessing proof assistant feedback for reinforcement learning and Monte-Carlo tree search*. arXiv: 2408.08152.

Xin, Huajian et al. (2024b). *DeepSeek-Prover: Advancing theorem proving in LLMs through large-scale synthetic data*. arXiv: 2405.14333.

Yang, Kaiyu et al. (2023). *LeanDojo: Theorem proving with retrieval-augmented language models*. arXiv: 2306.15626.

Yousefzadeh, Roozbeh, Xuenan Cao, and Azim Ospanov (2024). *A Lean dataset for International Math Olympiad: Small steps towards writing math proofs for hard problems*. arXiv: 2411.18872.

Zheng, Kunhao, Jesse Michael Han, and Stanislas Polu (2021). *MiniF2F: A cross-system benchmark for formal Olympiad-level mathematics*. arXiv: 2109.00110.